

Principles of Geographic Information Systems

MODULE-3: Spatial Referencing and Positioning, Data Entry and Preparation



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Certificate

This is to certify that the e-book titled “PRINCIPLES OF GEOGRAPHIC INFORMATION SYSTEMS” comprises all elementary learning tools for a better understating of the relevant concepts. This e-book is comprehensively compiled as per the predefined eight parameters and guidelines.



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T.Y. B.Sc. IT Sem-6

Principles of Geographic Information Systems

Unit III

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Spatial Referencing and Positioning

Spatial Referencing: Reference surfaces for mapping, Coordinate Systems, Map Projections, Coordinate Transformations,

Satellite-based Positioning: Absolute positioning, Errors in absolute positioning, Relative positioning, Network positioning, code versus phase measurements, Positioning technology.

Data Entry and Preparation

Spatial Data Input: Direct spatial data capture, Indirect spatial data capture, Obtaining spatial data elsewhere

Data Quality: Accuracy and Positioning, Positional accuracy, Attribute accuracy, temporal accuracy, Lineage, Completeness, Logical consistency

Data Preparation: Data checks and repairs, Combining data from multiple sources
Point Data Transformation: Interpolating discrete data, Interpolating continuous data

Recommended Books

1. Principles of Geographic Information Systems: An introductory textbook, Editors: Otto Huisman and Rolf A.
2. Principles of Geographical Information Systems – Peter A. Burrough (Author), Rachael A. McDonnell (Author), Christopher D. Lloyd (Author)

Prerequisites and Linking:

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Chapter 4

Spatial referencing and positioning

Objectives:

- i) To learn the relevance and actual use of reference surfaces, Coordinate systems and coordinate transformations.
- ii) To know about satellite-based positioning.
- iii) Introduction of global positioning techniques.

4. Spatial referencing and positioning

In the early days of GIS, users were mainly handling spatially referenced data from a single country. This data was usually derived from paper maps published by the country's mapping organization. Nowadays, GIS users are combining spatial data from a given country with global spatial data sets, reconciling spatial data from published maps with coordinates established with satellite positioning techniques and integrating their spatial data with that from neighboring countries. To perform these kinds of tasks successfully, GIS users need to understand basic spatial referencing concepts.

4.1 Spatial referencing

One of the defining features of GIS is their ability to combine spatially referenced data. A frequently occurring issue is the need to combine spatial data from different sources that use different spatial reference systems. This section provides a concept relating to the nature of spatial reference systems and the translation of data from one spatial referencing system into another.

4.1.1 Reference surfaces for mapping

The surface of the Earth is anything but uniform. The oceans can be treated as reasonably uniform, but the surface or topography of the land masses exhibits large vertical variations between mountains and valleys. These variations make it

impossible to approximate the shape of the Earth with any reasonably simple mathematical model. Consequently, two main reference surfaces have been established to approximate the shape of the Earth.

The Geoid and ellipsoid

One reference surface is called the Geoid, the other reference surface is the ellipsoid.

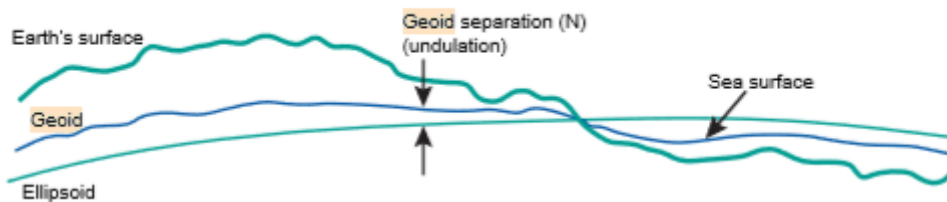


Figure 1: The Earth's surface, and two reference surfaces used to approximate it: the Geoid,

These are illustrated in Figure 1. Below, we look at and discuss the respective uses of each of these surfaces. And a reference ellipsoid. The Geoid separation (N) is the deviation between the Geoid and a reference ellipsoid

Video on Sea-level

Source: <https://www.youtube.com/watch?v=q65O3qA0-n4>

The Geoid and the vertical datum

We can simplify matters by imagining that the entire Earth's surface is covered by water. If we ignore tidal and current effects on this 'global ocean', the resultant water surface is affected only by gravity. This has an effect on the shape of this surface because the direction of gravity—more commonly known as plumb. Plumb line line—is dependent on the mass distribution inside the Earth. Due to irregularities or mass anomalies in this distribution the 'global ocean' results in an undulated surface. This surface is called the Geoid (Figure 2). The plumb line through any surface point is always perpendicular to it.

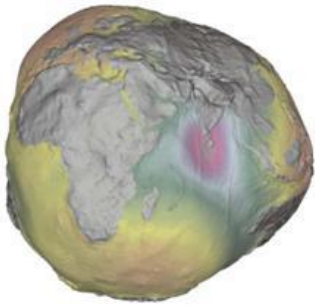


Figure 2: The Geoid, Exaggerated to illustrate the complexity of its surface.

The Geoid is used to describe heights. In order to establish the Geoid as reference for heights, the ocean's water level is registered at coastal places over several years using tide gauges (mareographs). Averaging the registrations largely eliminates variations of the sea level with time. The resulting water level represents an approximation to the Geoid and is called the mean sea level. For the Netherlands and Germany, the local mean sea level is realized through the Amsterdam tide-gauge (zero height). We can determine the height of a point in Enschede with respect to the Amsterdam tide gauge using a technique known as geodetic levelling (Figure 3). The result of process will be the height above local mean sea level for the Enschede point. The height determined with respect to a tide-gauge station is known as the orthometric height (height H above the Geoid).

Obviously, there are several realizations of local mean sea levels (also called local vertical datums) in the world. They are parallel to the Geoid but offset by up to a couple of meters. This offset is due to local phenomena such as ocean currents, tides, coastal winds, water temperature and salinity at the location of the tide

gauge. Care must be taken when using heights from another local vertical datum. For example, this might be the case in the border area of adjacent nations. Even within a country, heights may differ depending on to which tide gauge, mean sea level point, they are related. As an example, the mean sea level from the Atlantic to the Pacific coast of the USA increases by 0.6 to 0.7 m. The tide gauge (zero height) of the Netherlands differs -2.34 meters from the tide gauge (zero height) of the neighboring country Belgium. The local vertical datum is implemented through a levelling network (see Figure 3(a)). A levelling network consists of benchmarks, whose height above mean sea level has been determined through geodetic levelling. The implementation of the datum enables easy user access. The surveyors do not need to start from scratch (i.e. from the Amsterdam tide-gauge) every time they need to determine the height of a new point. They can use the benchmark of the levelling network that is closest to the point of interest (Figure 3(b)).

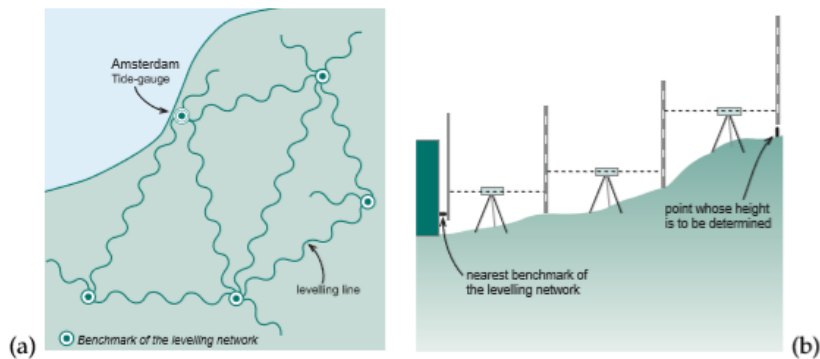


Figure 3: A levelling network implements a local vertical datum:

(a) network of levelling lines starting from the Amsterdam tide-gauge, showing some of the benchmarks; (b) how the orthometric height (H) is determined for some point, working from the nearest benchmark.

As a result of satellite gravity missions, it is currently possible to determine the height (H) above the Geoid with centimeter level accuracy. It is foreseeable that a global vertical datum may become ubiquitous in the next 10-15 years. If all published maps are also using this global vertical datum by that time, heights will become globally comparable, effectively making local vertical datums redundant for GIS users.

The ellipsoid

Above, we have defined a physical surface, the Geoid, as a reference surface for heights. We also need a reference surface for the description of the horizontal coordinates of points of interest. Since we will later project these horizontal coordinates onto a mapping plane, the reference surface for horizontal coordinates requires a mathematical definition and description. The most convenient geoOblate ellipsoid metric reference is the oblate ellipsoid (Figure 4). It provides a relatively simple figure which fits the Geoid to a first order approximation, though for small scale mapping purposes a sphere may be used. An ellipsoid is formed when an ellipse is rotated about its minor axis. This ellipse which defines an ellipsoid or spheroid is called a meridian ellipse.¹

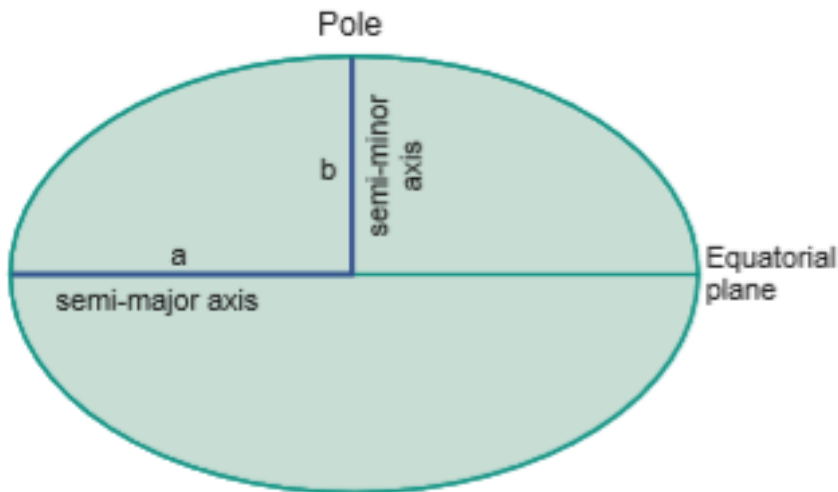


Figure 4: An oblate ellipse, defined by its semimajor axis a & semiminor axis b .

The shape of an ellipsoid may be defined in a number of ways, but in geodetic practice the definition is usually by its semi-major axis and flattening (Figure 4).

Flattening f is dependent on both semi-major axis a and the semi-minor axis b .

$$f = \frac{(a - b)}{a}.$$

The ellipsoid may also be defined by its semi-major axis a and its eccentricity e , which is given by:

$$e^2 = (1 - (\frac{b^2}{a^2})) = \frac{(a^2 - b^2)}{a^2} = 2f - f^2.$$

Given one axis and any one of the other three parameters, the other two can be derived. Typical values of the parameters for an ellipsoid are:

$$a = 6378135.00 \text{ m}, b = 6356750.52 \text{ m}, f = \frac{1}{298.26}, e = 0.08181881066$$

Many different ellipsoids have been defined. Local ellipsoids have been established to fit the **Geoid** (mean sea level) well over an area of local interest, which in the past was never larger than a continent. This meant that the differences between the **Geoid** and the reference ellipsoid could effectively be ignored, allowing accurate maps to be drawn in the vicinity of the datum (Figure 4.5).

With increasing demands for global surveying, work is underway to develop global reference ellipsoids. In contrast to local ellipsoids, which apply only to a

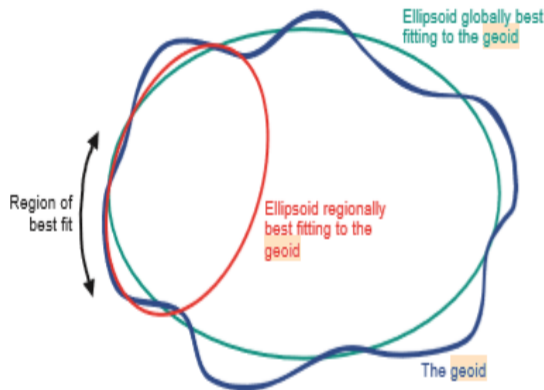


Figure 4.5: The **Geoid**, a globally best fitting ellipsoid for it, and a regionally best fitting ellipsoid for it, for a chosen region. Adapted from: Ordnance Survey of Great Britain. *A Guide to Coordinate Systems in Great Britain.*

The local horizontal datum

Ellipsoids have varying position and orientations. An ellipsoid is positioned and oriented with respect to the local mean sea level by adopting a latitude () and longitude () and ellipsoidal height (h) of a so-called fundamental point and an azimuth to an additional point. We say that this defines a local horizontal datum. Notice that the term horizontal datum and geodetic datum are being treated as equivalent and interchangeable words.

Several hundred local horizontal datums exist in the world. The reason is obvious: Different local ellipsoids with varying position and orientation had to be adopted to best fit the local mean sea level in different countries or regions. An example is the Potsdam Datum, the local horizontal datum used in Germany. The fundamental point is in Rauenberg and the underlying ellipsoid is the Bessel ellipsoid ($a = 6,377,397.156$ m, $b = 6,356,079.175$ m). We can determine the latitude and longitude (;) of any other point in Germany with respect to this local horizontal datum using geodetic positioning techniques, such as triangulation and trilateration. The result of this process will be the geographic (or horizontal) coordinates (;) of the new point in the Potsdam Datum.

Video on datum https://www.youtube.com/watch?v=xKGIMp_jog

Triangulation networks

A local horizontal datum is realized through a triangulation network. Such a network consists of monument points forming a network of triangular mesh elements (Figure 6). The angles in each triangle are measured in addition to

at least one side of a triangle; the fundamental point is also a point in the triangulation network. The angle measurements and the adopted coordinates of the Fundamental point are then used to derive geographic coordinates (;) for all monument points of the triangulation network.

Within this framework, users do not need to start from scratch (i.e. from the fundamental point) in order to determine the geographic coordinates of a new point. They can use the monument of the triangulation network that is closest to the new point. The extension and re-measurement of the network is nowadays done through satellite measurements.



Figure 6: The old primary triangulation network in the Netherlands made up of 77 points (mostly church towers). The extension and re-measurement of the network is nowadays done through satellite measurements. Adapted from original figure by 'Dutch Cadastre and Land Registers' now called het Kadaster.

The global horizontal datum

Local horizontal datums have been established to fit the Geoid well over the area of local interest, which in the past was never larger than a continent. With increasing demands for global surveying activities are underway to establish global reference surfaces. The motivation is to make geodetic results mutually comparable and to provide coherent results also to other disciplines like astronomy and geophysics.

The most important global (geocentric) spatial reference system for the GIS community is the International Terrestrial Reference System (ITRS). It is a three-dimensional coordinate system with a well-defined origin (the centre of mass ITRS of the Earth) and three orthogonal coordinate axes (X; Y; Z). The Z-axis points towards a mean Earth north pole. The X-axis is oriented towards a mean Greenwich meridian and is orthogonal to the Z-axis. The Y -axis completes the right-handed reference coordinate system (Figure 7a).

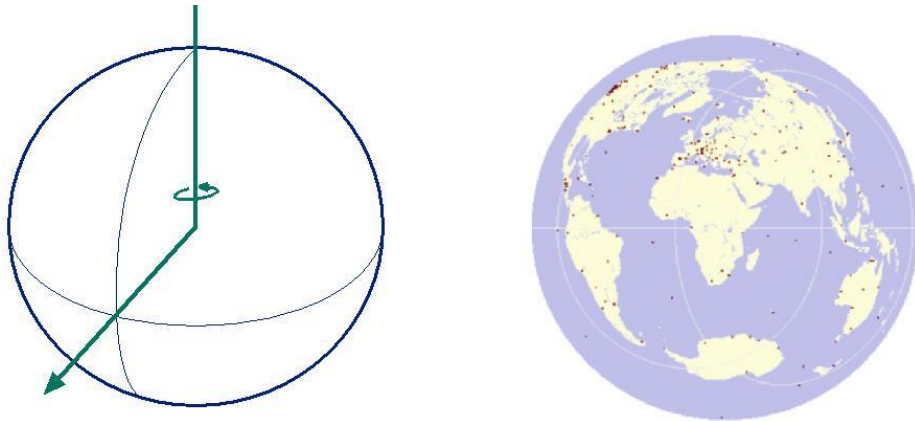


Figure 7: (a) The International Terrestrial Reference System (ITRS), and;- (b) the International Terrestrial Reference Frame (ITRF) visualized as a distributed set of ground control stations (represented by red points).

The ITRS is realized through the International Terrestrial Reference Frame (ITRF), a distributed set of ground control stations that measure their position continuously using GPS (Figure 7b).

Constant re-measuring is needed because of the involvement of new control stations and ongoing geophysical processes (mainly tectonic plate motion) that deform the Earth's crust at measurable global, re-gional and local scales. These deformations cause positional differences in time, and have resulted in more than one realization of the ITRS. Examples are the ITRF ITRF96 or the ITRF2000. The ITRF96 was established at the 1st of January, 1997.

This means that the measurements use data up to 1996 to fix the geocentric coordinates (X, Y and Z in metres) and velocities (positional change in X, Y and Z in metres per year) at the different stations. The velocities are used to propagate the measurements to other epochs (times). The trend is to use the ITRF everywhere in the world for reasons of global compatibility.

GPS uses the World Geodetic System 1984 (WGS84) as its reference system. It has been refined on several occasions and is now aligned with the ITRF to within a few centimetres worldwide. Global horizontal datums, such as the ITRF2000 or WGS84, are also called geocentric datums because they are geocentrically Geocentric datums with respect to the centre of mass of the Earth. They became available

Since the size and shape of satellite orbits is directly related to the centre of mass of the Earth, observations of natural or artificial satellites can be used to pinpoint the centre of mass of the Earth, and hence the origin of the ITRS.³ This technique can also be used for the realization of the global ellipsoids and datums at the accuracy level required for large-scale mapping.

We can easily transform ITRF coordinates (X, Y and Z in metres) into geo-graphic coordinates with respect to the GRS80 ellipsoid without the loss of accuracy. However, the ellipsoidal height h , obtained through this straight-forward transformation, has no physical meaning and does not correspond to intuitive human perception of height. We therefore use the height H , above 3D spatial referencing the Geoid (see Figure 8). It is foreseeable that global 3D spatial referencing, in terms of (λ ; ϕ ; H), could become ubiquitous in the next 10–15 years. If all published maps are also globally referenced by that time, the underlying spatial referencing concepts will become transparent and hence redundant for GIS users.

Video on Coordinate System Jargon- geoid, datum, projection
Source:https://www.youtube.com/watch?v=Z41Dt7_R180

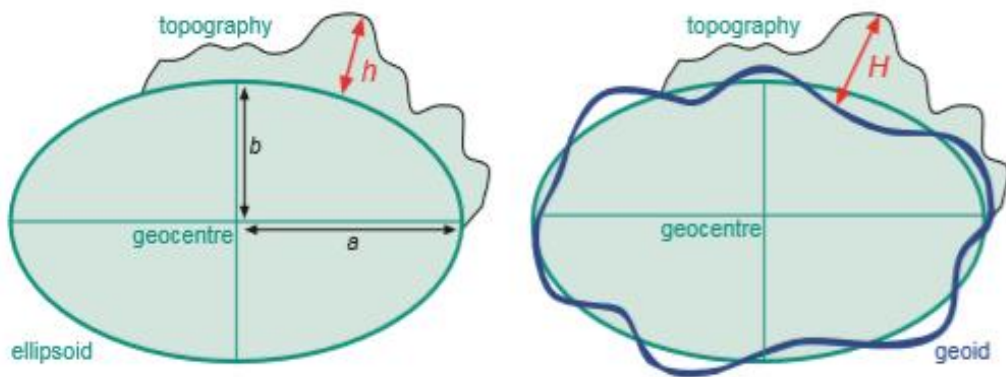


Figure 8: Height h above the geocentric ellipsoid, and height H above the Geoid. The first is measured orthogonal to the ellipsoid, the second orthogonal to the Geoid.

Hundreds of existing local horizontal and vertical datums are still relevant because they form the basis of map products all over the world. For the next few years we will be required to deal with both local and global datums until the former are eventually phased out. During the transition period, we will require tools to transform coordinates from local horizontal datums to a global horizontal datum and vice versa.

Extra-terrestrial positioning techniques include Satellite Laser Ranging (SLR), Lunar Laser Ranging (LLR), Global Positioning System (GPS), and Very Long Baseline Interferometry (VLBI), among others.

The organizations that usually develop transformation tools and make them available to the user community are provincial or National Mapping Organizations (NMOs) and cadastral authorities.

4.1.2 Coordinate systems

As mentioned before, the special nature of spatial data obviously lies in it being spatially referenced. Different kinds of coordinate systems are used to position data in space. Here we distinguish between spatial and planar coordinate systems. Spatial (or global) coordinate systems are used to locate data either on the Earth's surface in a 3D space, or on the Earth's reference surface (ellipsoid or sphere) in a 2D space. Below we discuss the geographic coordinate system in a 2D and 3D space and the geocentric coordinate system, also known as the 3D Cartesian coordinate system. Planar coordinate systems on the other hand are used to locate data on the flat surface of the map in a 2D space. We will discuss the 2D Cartesian coordinate system and the 2D polar coordinate system.

2D Geographic coordinates (;

The most widely used global coordinate system consists of lines of geographic latitude (ϕ or ϕ°) and longitude (λ or λ°). Lines of equal latitude are called parallels. They form circles on the surface of the ellipsoid⁴. Lines of equal longitude are called meridians and they form ellipses (meridian ellipses) on the ellipsoid. (Figure 9)

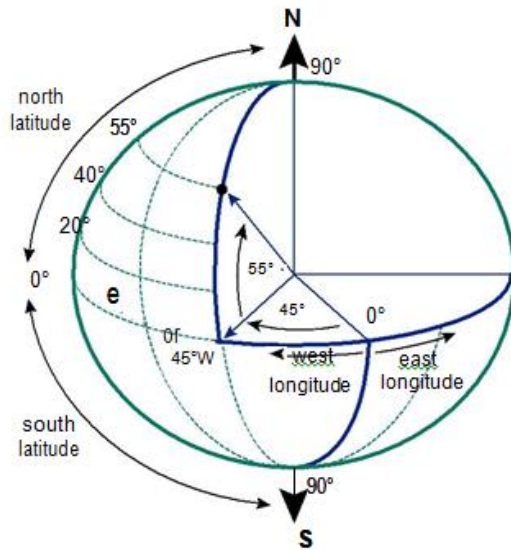


Figure 9: The latitude () and longitude () angles represent the 2D geographic coordinate system

normal through P' and the equatorial plane. Latitude is zero on the equator (= 0), and increases towards the two poles to maximum values of = +90 (N 90) at the North Pole and = -90 (S 90) at the South Pole.

The longitude () is the angle between the meridian ellipse which passes through Greenwich and the meridian ellipse containing the point in question. It is measured in the equatorial plane from the meridian of Greenwich (= 0) either eastwards through = + 180 (E 180) or westwards through = -180 (W 180)

Latitude and longitude represent the geographic coordinates (;) of a point P' (Figure 10) with respect to the selected reference surface. They are always given in angular units. For example, the coordinates for City hall in Enschede are:⁵

= 52 13⁰26:2⁰⁰N; = 6 53⁰32:1⁰⁰E

The graticule on a map represents the projected position of the geographic coordinates (;) at constant intervals, or in other words the projected position of selected meridians and parallels (Figure 13). The shape of the graticule depends largely on the characteristics of the map projection and the scale of the map.

3D Geographic coordinates

3D geographic coordinates are obtained by introducing the ellipsoidal height h to the system. The ellipsoidal height (h) of a point is the vertical distance of the point in question above the ellipsoid. It is measured in distance units along the ellipsoidal normal from the point to the ellipsoid surface. 3D geographic coordinates can be used to define a position on the surface of the Earth.

Video 1: Coordinate System Jargon- geoid, datum, projection

Source:https://www.youtube.com/watch?v=Z41Dt7_R180

4.1.3 Map projections

Maps are one of the world's oldest types of document. For quite some time it was thought that our planet was flat, and during those days, a map simply was a miniature representation of a part of the world. Now that we know that the Earth's surface is curved in a specific way, we know that a map is in fact a flattened representation of some part of the planet. The field of map projections concerns itself with the ways of translating the curved surface of the Earth into a flat map.

A map projection is a mathematically described technique of how to represent the Earth's curved surface on a flat map

To represent parts of the surface of the Earth on a flat paper map or on a computer screen, the curved surface, the curved horizontal reference surface must be mapped

onto the 2D mapping plane. The reference surface for large scale mapping is usually, an oblate ellipsoid and for small scale mapping a sphere. Mapping onto a 2D mapping plane means transforming each point on the reference surface with geographic coordinates to a set of cartesian coordinates (x, y) representing position on the map plane (Figure 15).

Active Map Projection <https://www.youtube.com/watch?v=z1PpoolrMK4>

<https://www.youtube.com/watch?v=gGumy-9HrSY>

Classification of map projections

These qualities in turn make resulting maps useful for certain purposes. By definition, any map projection is associated with scale distortions. Hundreds of map projections have been developed, each with its own specific characteristics. There is simply no way to flatten out a piece of ellipsoidal or spherical surface without stretching some parts of the surface more than others. The amount and which kind of distortions a map will have depends on the type of the map projection that has been selected.

Typical choices for such intermediate surfaces are cones and cylinders. Such map projections are then called conical, and cylindrical, respectively. Figure 16 shows the surfaces involved in these three classes of projections.

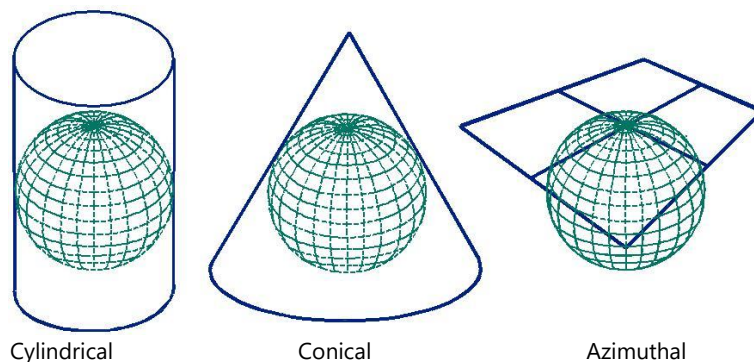


Figure 16: Classes of map projections

In the geometric depiction of map projections in Figures 16 and 17, the symmetry axes of the plane, cone and cylinder coincide with the rotation axis of the ellipsoid or sphere, i.e. a line through N and S pole. In this case, the projection is said to be a normal projection. The other cases are transverse projections (symmetry axis in the equator) and oblique projections (symmetry axis is somewhere between the rotation axis and equator of the ellipsoid or sphere). These cases are illustrated in Figure 18.

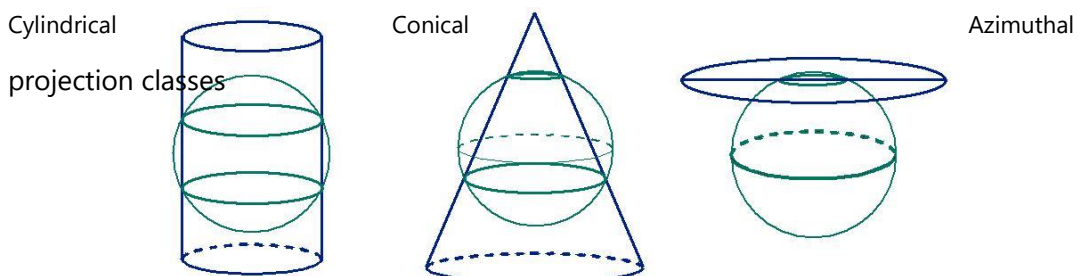


Figure 17: Three secant

The Universal Transverse Mercator (UTM) uses a transverse cylinder, secant to the horizontal reference surface. UTM is an important projection used world-wide. The projection is a derivation from the Transverse Mercator projection (also known as Gauss-Kruger or Gauss conformal projection). The UTM divides the world into 60 narrow longitudinal zones of 6 degrees, numbered from 1 to 60. The narrow zones of 6 degrees (and the secant map surface) make the distortions small enough for large scale topographic mapping.

Normal cylindrical projections are typically used to map the world in its entirety. Conical projections are often used to map the different continents, while the normal azimuthal projection may be used to map the polar areas. Transverse and oblique aspects of many projections can be used for most parts of the world.

The planar, conical, and cylindrical surfaces in Figure 16 are all tangent surfaces they touch the horizontal reference surface in one point (plane) or along a closed line (cone and cylinder only). Another class of projections is obtained if the surfaces are chosen to be secant to (to intersect with) the horizontal reference surface; illustrations are in Figure 17. Then, the reference surface is intersected along one closed line or two closed lines (cone and cylinder) Secant map surfaces are used to reduce or average out scale errors because the line of intersection are not distorted on the map,

4.1.4 Coordinate transformations

Map and GIS users are mostly confronted in their work with transformations from one two-dimensional coordinate system to another. This includes the transformation of polar coordinates delivered by the surveyor into Cartesian map coordinates or the transformation from one 2D Cartesian $(x; y)$ system of a specific map projection into another 2D Cartesian $(x^0; y^0)$ system of a defined map projection.

Datum transformations are transformations from a 3D coordinate system (i.e. horizontal datum) into another 3D coordinate system. These kinds of

transformations are also important for map and GIS users. They are usually collecting spatial data in the field using satellite navigation technology and need to represent this data on published map on a local horizontal datum.

We may relate an unknown coordinate system to a known coordinate system on the basis of a set of selected points whose coordinates are known in both systems. These points may be ground control points (GCPs) or common points such as corners of houses or road intersections, as long as they have known coordinates in both systems. Image and scanned data are usually transformed by this method. The transformations may be conformal, affine, polynomial, or of another type, depending on the geometric errors in the data set.

2D Polar to 2D Cartesian transformations

The transformation of polar coordinates ($r; \theta$), into Cartesian map coordinates ($x; y$) is done when field measurements, angular and distance measurements are transformed into map coordinates.

Changing map projection

Forward and inverse mapping equations are normally used to transform data from one map projection to another. The inverse equation of the source projection is used first to transform source projection coordinates ($x; y$) to geographic coordinates ($\lambda; \phi$). Next, the forward equation of the target projection is used to transform the geographic coordinates ($\lambda; \phi$) into target projection coordinates ($x^0; y^0$). The first equation takes us from a projection A into geographic coordinates. The second takes us from geographic coordinates ($\lambda; \phi$) to another map projection B. These principles are illustrated in Figure 22.

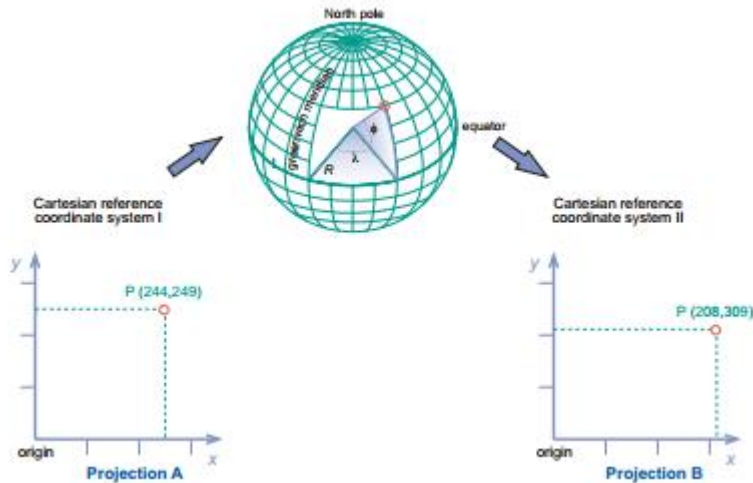


Figure 22: The Principle of changing from one map projection into another

Datum transformations

A change of map projection may also include a change of the horizontal datum. This is the case when the source projection is based upon a different horizontal datum than the target projection. If the difference in horizontal datums is ignored, there will not be a perfect match between adjacent maps of neighbouring countries or between overlaid maps originating from different projections. It may result in up to several hundred meters difference in the resulting coordinates. Therefore, spatial data with different underlying horizontal datums may need a so-called datum transformation.

Suppose we wish to transform spatial data from the UTM projection to the Dutch RD system, and that the data in the UTM system are related to the European Datum 1950 (ED50), while the Dutch RD system is based on the Amersfoort datum. In this example the change of map projection should be combined with a datum transformation step for a perfect match. This is illustrated in Figure 23.

The inverse equation of projection A is used first to take us from the map coordinates (x ; y) of projection A to the geographic coordinates (λ ; ϕ ; h) in datum A. A height coordinate (h or H) may be added to the (x ; y) map coordinates. Next, the datum transformation takes us from these coordinates to the geographic coordinates (λ ; ϕ ; h) in datum B. Finally, the forward equation of projection B is used to take us from the geographic coordinates (λ ; ϕ ; h) in datum B to the map

coordinates $(x^0; y^0)$ of projection B.

Mathematically a datum transformation is feasible via the geocentric coordinates $(x; y; z)$, or directly by relating the geographic coordinates of both datum systems. The latter relates the ellipsoidal latitude (ϕ) and longitude (λ) , and

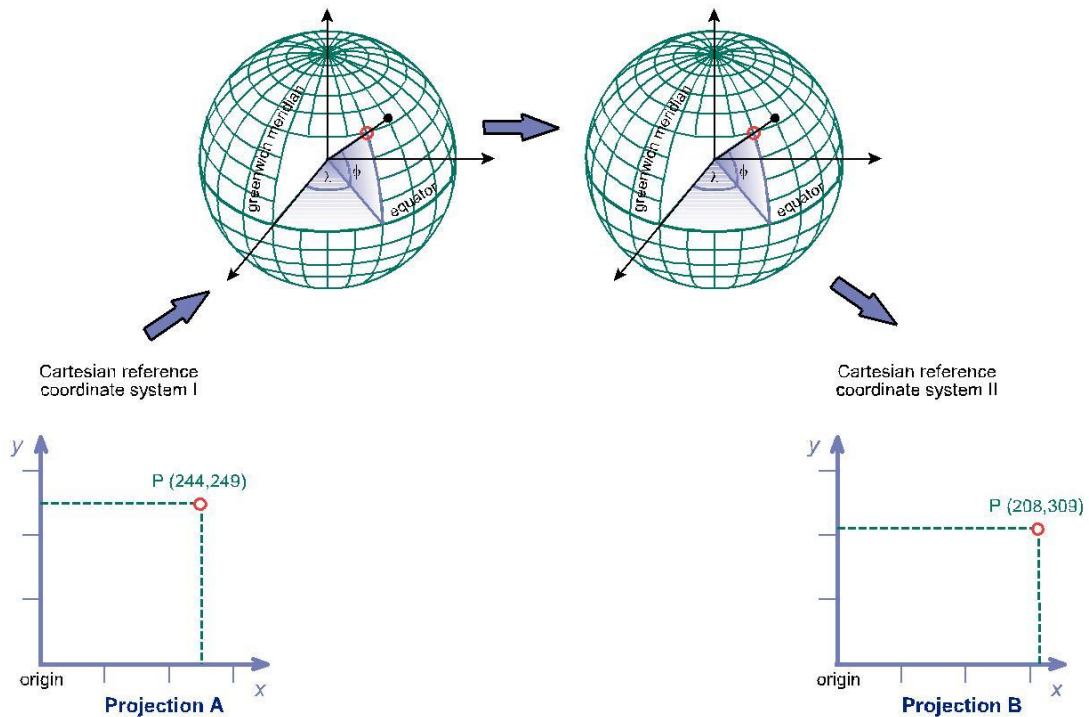


Figure 23: The principle of changing from one projection into another, combined with a datum transformation from datum A to datum B

4.2 Satellite-based positioning

The previous section has noted the importance of satellites in spatial referencing. Satellites have allowed us to realize geocentric reference systems, and increase the level of spatial accuracy substantially. They are critical tools in geodetic engineering for the maintenance of the ITRF. They also play a key role in mapping, surveying, and in a growing number of applications requiring positioning techniques. Nowadays, for fieldwork that includes spatial data acquisition, the use of satellite-based positioning is considered indispensable.

Satellite-based positioning was developed and implemented to address military needs, somewhat analogously to the early development of the internet. The technology is now widely available for civilians use. The requirements for the development of the positioning system were:

- i) Suitability for all kinds of military use: ground troops and vehicles, aircraft and missiles, ships;
- ii) Requiring only low-cost equipment with low energy consumption at the receiver end;
- iii) Provision of results in real time for an unlimited number of users concurrently;

Support for different levels of accuracy (military versus civilian); Around-the-clock and weather-proof availability; Use of a single geodetic datum;

Protection against intentional and unintentional disturbance, for instance, through a design allowing for redundancy.

A satellite-based positioning system set-up involves implementation of three hardware segments:

- i) The **space segment**, i.e. the satellites that orbit the Earth, and the radio signals that they emit,
- ii) The **control segment**, i.e. the ground stations that monitor and maintain the space segment components, and
- iii) The **user segment**, i.e. the users with their hard and software to conduct positioning.

In satellite positioning, the central problem is to determine values (X , Y , Z) of a receiver that receives satellite signals, i.e. to determine the position of the receiver with a stated accuracy and precision. Required accuracy and precision depends on the application; timeliness, i.e. are the position values required in real time or can they be determined later during post processing, also varies between applications. Finally, some applications like navigation require kinematic approaches, which take into account the fact that the receiver is not stationary, but is moving.

Video on Satellite Positioning

https://www.youtube.com/watch?v=r0r4P1UAv_g

4.2.1 Absolute positioning

The working principles of absolute, satellite-based positioning are fairly simple:

1. A satellite, equipped with a clock, at a specific moment sends a radio message that includes the *satellite identifier*,

its *position in orbit*, and its *clock reading*.

2. A receiver on or above the planet, also equipped with a clock, receives the message slightly later, and reads its own clock.

3. From the time delay observed between the two clock readings, and knowing

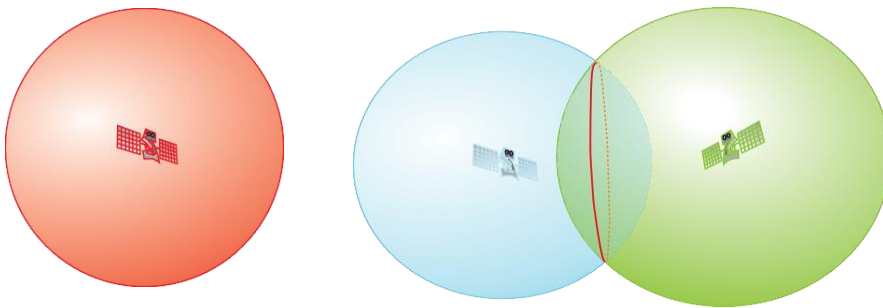
The *pseudorange* of a satellite with respect to a receiver, is its apparent distance to the receiver, computed from the time delay with which its radio signal is received.

the speed of radio transmission through the medium between (satellite) sender and receiver, the receiver can compute the distance to the sender, also known as the satellite's *pseudorange*. such a computation determines the position of the receiver to be on a sphere of radius equal to the computed pseudorange (refer to Figure 4.24(a)). If the receiver instantaneously would do the same with a message of another satellite that is positioned elsewhere, the position of the receiver is restricted to another sphere. The intersection of the two spheres, which have different centres, determines a

circle as the set of possible positions of the receiver (refer to Figure 24(b)). If a third satellite message is taken into consideration, the inter

Trilateration section of three spheres determines at most two positions, one of which is the actual position of the receiver. In most, if not all, practical situations where two positions result, one of them is a highly unlikely position for a signal receiver. The overall procedure is known as *trilateration*: the determination of a position based on three distances. clock that is not synchronized with the satellite clocks. This brings into play an additional unknown parameter, namely the synchronization bias of the receiver clock, i.e. the difference in time reading between it and the satellite clocks.

Our set of unknown variables has now become $(X, Y, Z, \Delta t)$ representing a 3D position and a clock bias. By including the information obtained from a fourth satellite message, we can solve the problem (see Figure 4.25).



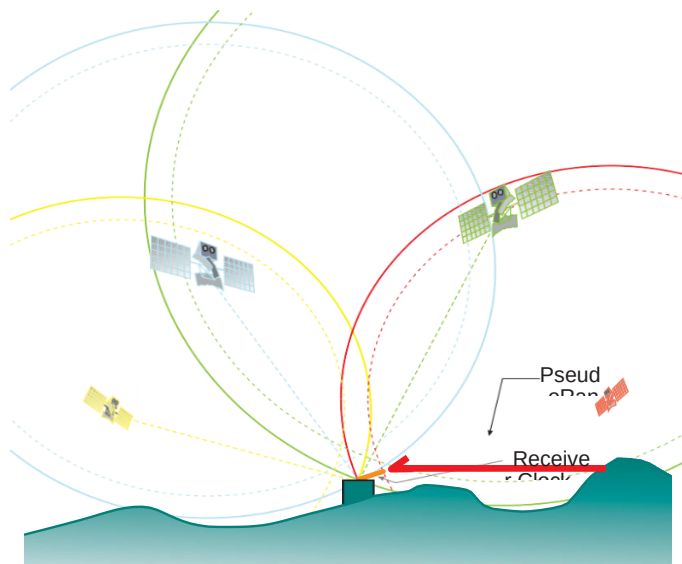
b) This will result in the determination of the receiver's actual position (X, Y, Z) , as well as its receiver clock bias Δt , and if we correct the receiver clock for this bias we effectively turn it into a high precision, atomic clock as well!

Obtaining a high precision clock is a fortunate side-effect of using the receiver, as it allows the design of experiments distributed in geographic space that demand high levels of synchrony. One such application is the use of wireless sensor networks for various natural phenomena like earthquakes, meteorological patterns or in water management.

Another application is in the positioning of mobile phone users making an emergency call. Often the caller does not know their location accurately. The telephone company can trace back the call to the receiving transmitter mast, but this may be servicing an area with a radius of 300 m to 6 km. That is too inaccurate a position for an emergency ambulance to go to. However, if all masts in the telephony network are equipped with a satellite positioning receiver (and thus, with a very good, synchronized clock) the time of reception of the call at each

mast can be recorded. The *time difference of arrival* of the call between two nearby masts determines a hyperbola on the ground of possible positions of the caller; if the call is received on three masts, we would have two hyperbolas, allowing intersection, and thus 'hyperbolic positioning'. With current technology the (horizontal) accuracy would be better than 30 m. Returning to the subject of satellite-based positioning, when only three and not four satellites are 'in view', the receiver is capable of falling back from the above *3D positioning mode* to the inferior *2D positioning mode*. With the relative abundant 2D positioning mode dance of satellites in orbit around the earth, this is a relatively rare situation, but it serves to illustrate the importance of 3D positioning.

If a 3D fix has already been obtained, the receiver simply assumes that the height above the ellipsoid has not changed since the last 3D fix. If no fix had yet been obtained, the receiver assumes that it is positioned at the geocentric ellipsoid adopted by the positioning system, i.e. at height $h=0$.⁸ In the receiver computations, the ellipsoid fills the slot of the missing fourth satellite sphere, and the unknown variables can therefore still be determined. Clearly in both of these cases, the assumption for this computation is flawed and the positioning results in 2D mode will be unreliable—much more so if no previous fix had been obtained and one's receiver is not at all near the surface of the geocentric ellipsoid.



Time, clocks and world time

When trains became an important means of transportation, these local time systems became problematic as the schedules required a single time system. Such a time system needed the definition of *time zones*: typically as 24 geographic strips certain longitudes that are multiples of 15° . This all gave rise to Greenwich Mean Time (GMT). GMT was the world time standard of choice. It was a system based on the mean solar time at the meridian of Greenwich, United Kingdom, which is the conventional 0-meridian in geography.

GMT was later replaced by Universal Time (UT), a system still based on meridian crossings of stars, but now of far away quasars as this provides more accuracy than that of the Sun. It is still the case that the rotational velocity of our planet is not constant and the length of a solar day is increasing. So UT is not a perfect system either. It continues to be used for civil clock time, but it is officially now replaced by International Atomic Time (TAI). UT actually has various versions, amongst which are UT0, UT1 and UTC. UT0 is the Earth rotational time observed in some location. Because the Earth experiences polar motion as well, UT0 differs between locations. If we correct for polar motion, we obtain UT1, which is identical everywhere. It is still a somewhat erratic clock because of the earlier mentioned varying rotational velocity of the planet. The uncertainty is about 3 msec per day.

Coordinated Universal Time (UTC) is used in satellite positioning, and is maintained with atomic clocks. By convention, it is always within a margin of 0.9 sec of UT1, and twice annually it may be given a shift to stay within that margin. This occasional shift of a *leap second* is applied at the end of June 30 or preferably at the end of December 31. The last minute of such a day is then either 59 or

UTC 61 seconds long. So far, adjustments have always been to add a second. UTC time can only be determined to the highest precision after the fact, as atomic time is determined by the reconciliation of the observed differences between a number of atomic clocks maintained by different national time bureaus.

In recent years we have learned to measure distance, therefore also position, with clocks using satellite signals. The conversion factor is the speed of light, approximately $3 \cdot 10^8$ m/s in vacuum. No longer can multiple seconds of clock bias be allowed, and this is where atomic clocks come in. They are very accurate

Atomic clocks timekeepers, based on the exactly known frequency with which specific atoms (Cesium, Rubidium and Hydrogen) make discrete energy state

jumps. Positioning satellites usually have multiple clocks on board; ground control stations have even better quality atomic clocks.

VIDEO ON Atomic Clock

<https://www.youtube.com/watch?v=9ikbD7UGzol>

4.2.2 Errors in absolute positioning

For background information on the calculation of positional error (specifically, the calculation of RMSE or *root mean square error*)

Errors related to the space segment

As a first source of error, the operators of the control segment may intentionally deteriorate radio signals of the satellites to the general public, to avoid optimal use of the system by the enemy, for instance in times of global political tension and war. This *selective availability*—meaning that the military forces allied with the control segment *will* still have access to undisturbed signals—may cause error that is an order of magnitude larger than all other error sources combined.⁹ Secondly, the satellite message may contain incorrect information. Assuming that it will always know its own identifier, the satellite may make two kinds of error:

1. **Incorrect clock reading:** Even atomic clocks can be off by a small margin, and since Einstein, we know that travelling clocks are slower than resident clocks, due to a so-called relativistic effect. If one understands that a clock that is off

by 0.000001 sec causes a computation error in the satellite's pseudorange of approximately 300 m, it is clear that these satellite clocks require very strict monitoring.

2. ***Incorrect orbit position:*** The orbit of a satellite around our planet is easy to describe mathematically if both bodies are considered point masses, but in real life they are not. For the same reasons that the Geoid is not a simply shaped surface, the Earth's gravitation field that a satellite experiences

Both types of error are strictly monitored by the ground control segment, which is responsible for correcting any errors of this nature, but it does so by applying an agreed upon tolerance. A control station can obviously compare results of positioning computations like discussed above with its accurately *known* position, flagging any unacceptable errors, and potentially labelling a satellite as temporarily 'unhealthy' until errors have been corrected, and brought to within the tolerance. This may be done by uploading a correction on the clock or orbit settings to the satellite.

Errors related to the medium

Thirdly, the *medium* between sender and receiver may be of influence to the radio signals. The middle atmospheric layers of stratoand mesosphere are relatively harmless and of little hindrance to radio waves, but this is not true of the lower and upper layer. They are, respectively:

The troposphere:

the approximate 14 km high airspace just above the Earth's surface, which holds much of the atmosphere's oxygen and which envelopes all phenomena that we call the weather. It is an obstacle that delays radio waves in a rather variable way.

The ionosphere:

the most outward part of the atmosphere that starts at an altitude of 90 km, holding many electrically charged atoms, thereby forming a protection against various forms of radiation from space, including to some extent radio waves. The degree of ionization shows a distinct night and day rhythm, and also depends on solar activity. The latter is a more severe source of delay to satellite signals, which obviously means that pseudo ranges are estimated larger than they actually are.

When satellites emit radio signals at two or more frequencies, an estimate can be computed from differences in delay incurred for signals of different frequency, and this will allow for the correction of atmospheric delay, leading to a 10–50% improvement of accuracy. If this is not the case, or if the receiver is capable of receiving just a single frequency, a model should be applied to forecast the (especially ionospheric) delay, typically taking into account the time of day and current latitude of the receiver.

Errors related to the receiver's environment

Fourth in this list is the error occurring when a radio signal is received via two or more paths between sender and receiver, some of which typically via a bounce off of some nearby surface, like a building or rock face. The term applied to this Multi-path error phenomenon is *multi-path*; when it occurs the multiple receptions of the same signal may interfere with each other (see Figure 26). Multi-path is a difficult to avoid error source.

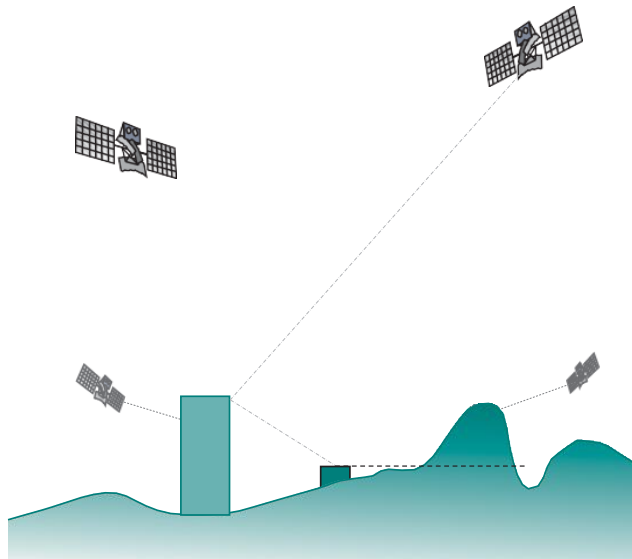


Figure 26: At any point in time, a number of satellites will be above the receiver's horizon. But not all of them will be 'in view' (like the left and right satellites), and for others multipath signal reception may occur.

All of the above error sources have an influence on the computation of a

satellite's pseudo range. In accumulation, they are called the *user equivalent range error* (UERE). Some error sources may be at work for all satellites being used by the receiver, for instance, selective availability and the atmospheric delay, while others may be specific to one satellite, for instance, incorrect satellite information and multi-path.

Errors related to the relative geometry of satellites and receiver

There is one more source of error that is unrelated to individual radio signal characteristics, but that rather depends on the combination of the satellite signals used for positioning. Of importance is their constellation in the sky from the receiver perspective. Referring to Figure 27, one will understand that the sphere intersection technique of positioning will provide more precise results when the four satellites are nicely spread over the sky, and thus that the satellite constellation of Figure 27(b) is preferred over the one of 27(a).

This error source is known as **geometric dilution of precision (GDOP)**. GDOP is lower when Geometric dilution of satellites are just above the horizon in mutually opposed compass directions. However, such satellite positions have bad atmospheric delay characteristics, so in practice it is better if they are at least 15° above the horizon. When more than four satellites are in view, modern receivers use ‘least-squares’ adjustment to calculate the best positional fix possible from all of the signals. This gives a better solution that just using the “best four”, as was done previously.

satellite clock	2.0 m
satellite position ionospheric delay	2.5 m
tropospheric delay receiver noise multi-path	0.5 m
	0.3 m
	0.5 m
Total RMSE Range error: $\sqrt{2^2 + 2.5^2 + 5^2 + 0.5^2 + 0.3^2 + 0.5^2} =$	5.97 m

Table 4.4: Indication of typical magnitude of errors in absolute satellite-based positioning precision values

These errors are not all of similar magnitude. An overview of some typical values

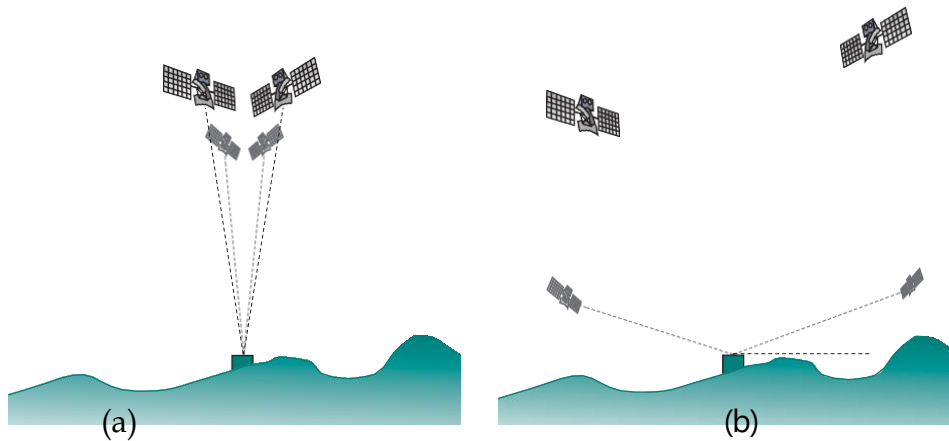


Figure 27: Geometric dilution of precision. The four satellites used for positioning can be in a bad constellation (a) or in a better constellation (b).

Figure 27: (without selective availability) is provided in Table 4.4. GDOP functions not so much as an independent error source but rather as a multiplying factor, decreasing the precision of position and time values obtained.

The procedure that we discussed above is known as *absolute, single-point positioning based on code measurement*. It is the fastest and simplest, yet least accurate way of determining a position using satellites. It suffices for recreational purposes and other applications that require horizontal accuracy not under 5–10 m. Typically, when encrypted military signals can also be used, on a dual-frequency receiver the achievable horizontal accuracy is 2–5 m. Below, we discuss other satellite-based positioning techniques with better accuracies.

4.2.3 Relative positioning

One technique of trying to remove errors from positioning computations is to perform many position computations, and to determine the average over the solutions. Many receivers allow the user to do so. It should however be clear from the above that *averaging* may address *random* errors like signal noise, selective availability (SA) and multi-path to some extent, but not

systematic sources of error, like incorrect satellite data, atmospheric delays, and GDOP effects.¹⁰

Random and systematic

These sources should be removed before averaging is applied. It has been shown that averaging over 60 minutes in absolute, single-point positioning based on code measurements, before systematic error removal, leads only to a 10–20% improvement of accuracy. In such cases, receiver averaging is therefore of limited value, and requires long periods under near-optimal conditions. Averaging is a good technique if systematic errors have been accounted for.

In relative positioning, also known as *differential positioning*, one tries to remove some of the systematic error sources by taking into account measurements of these errors in a nearby stationary *reference receiver* with an accurately known position. By using these systematic error findings at the reference, the position of the *target receiver* of interest will become known much more precisely.

In an optimal setting, reference and target receiver experience identical conditions and are connected by a direct data link, allowing the target to receive correctional data from the reference. In practice, relative positioning allows reference and target receiver to 70–200 km apart, and they will essentially experience similar atmospheric signal error. For each satellite in view, the reference receiver.

Finally, there is also a notion of *inverted relative positioning*. The principles are still as above, but in this technique the target receiver does not correct for satellite pseudorange error either, but uses a data link to upload its positioning/timing information to a central repository, where the corrections are applied. This can be useful in cases where many target receivers are needed and budget does not allow them to be expensive.

Video on [Absolute Location Versus Relative Location - YouTube](#)

4.2.4 Network positioning

After discussing the advantages of relative positioning, we can move on to the notion of *network positioning*: an integrated, systematic network of reference receivers covering a large area like a continent or even the whole globe.

The organization of such a network can take different shapes, augmenting an already existing satellite-based system. Here we discuss a general architecture, consisting of a network of *reference stations*, strategically positioned in the area to be covered, each of which is constantly monitoring signals and their errors for all positioning satellites in view. One or more *control centres* receive the reference station data, verify this for correctness, and relay (uplink) this information to a *geostationary satellite*. The satellite will retransmit the correctional data to the area that it covers, so that *target receivers*, using their own approximate position, determine satellite signal error correction, further accurate position fixes.

With network positioning, accuracy in the submetre range can be obtained. Typically, advanced receivers are required, but the technology lends itself also for solutions with a single advanced receiver that functions in the direct neighbourhood as a reference receiver to simple ones.

Video on Network Positioning

[What is Galileo? - YouTube](#)

4.2.5 Code versus phase measurements

Up until this point, we have assumed that the receiver determines the range of a satellite by measuring time delay on the received ranging code. There exists a more advanced range determination technique known as *carrier phase measurement*. This typically requires more advanced receiver technology, and longer observation sessions. Carrier phase measurement can currently only be used with relative positioning, as absolute positioning using this method is notll developed.

The technique aims to determine the number of cycles of the (sine-shaped) radio signal between sender and receiver. Each cycle corresponds to one wavelength of the signal, which in the applied L-band frequencies is 19–24 cm. Since this number of cycles cannot be directly measured, it is determined, in a long observation session, from the change in carrier phase with time. This happens because the satellite is orbiting itself. From its orbit parameters and the change in phase over time, the number of cycles can be derived.

With relative positioning techniques, a horizontal accuracy of 2 mm–2 cm can be achieved. This degree of accuracy makes it possible to measure tectonic plate movements, which can be as big as 10 cm per year in some locations on the planet.

4.2.6 Positioning technology

We include this section to provide the reader with a little information on currently available satellite-based positioning technology. It should be noted that this textbook will easily outlive the currency of the information

contained within it, as our technology is constantly evolving.

At present, two satellite-based positioning systems are operational (GPS and GLONASS), and a third is in the implementation phase (Galileo). Respectively, these are American, Russian and European systems. Any of these, but especially GPS and Galileo, will be improved over time, and will be augmented with new techniques.

GPS

The NAVSTAR Global Positioning System (GPS) was declared operational in 1994, providing Precise Positioning Services (PPS) to US and allied military forces as well as US government agencies, and Standard Positioning Services (SPS) to civilians throughout the world. Its space segment nominally consists of 24 satellites, each of which orbit our planet in 11h58m at an altitude of 20,200 km. There can be any number of satellites active, typically between 21 and 27. The satellite-Orbital planes are organized in six orbital planes, somewhat irregularly spaced, with an angle of inclination of $55\text{--}63^\circ$ with the equatorial plane, nominally having four satellites each (see Figure 4.28). This means that a receiver on Earth will have between five and eight (sometimes up to twelve) satellites in view at any point in time. Software packages exist to help in planning GPS surveys, identifying expected satellite set-up for any location and time.

The NAVSTAR satellites transmit two radio signals, namely the L1 frequency at 1575.42 MHz and the L2 frequency at 1227.60 MHz. The first two signals consist of

The role of the ranging codes is two-fold:

The carrier waves at the given frequencies, A coarse ranging code, known as C/A, modulated on L1,;A navigation message modulated on both L1 and L2.The role of L2 is to provide a second radio signal, thereby allowing (the more expensive) dual-frequency receivers a way of determining fairly precisely the actual ionospheric delay on satellite signals received.

An encrypted precision ranging code, known as P(Y), modulated on L1 and L2, and

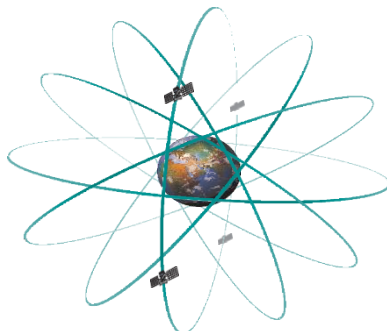


Figure 28: Constellation of satellites, four shown in only one orbit plane, in the GPS system.

To identify the satellite that sent the signal, as each satellite sends unique codes, and the receiver has a look-up table for these codes, and
To determine the signal transit time, and thus the satellite's pseudorange.

The navigation message contains the satellite orbit and satellite clock error information, as well as some general system information. GPS also carries a fifth, encrypted military signal carrying the M-code. GPS uses WGS84 as its reference system. It has been refined on several occasions and is now aligned with the WGS84 and ITRF. ITRF at the level of a few centimetres worldwide. GPS has adopted UTC as its time system.

In the civil market, GPS receivers of varying quality are available, their quality depending on the embedded positioning features: supporting single or dual frequency, supporting only absolute or also relative positioning, performing code measurements or also carrier phase measurements. Leica and Trimble are two of the well-known brands in the high-precision, professional surveying domain; Magellan and Garmin, for instance, operate in the lower price, higher volume consumer market range, amongst others for recreational use in outdoor activities. Many of these are single frequency receivers, doing only code measurements, though some are capable of relative positioning. This includes the new generation of GPS-enabled mobile phones.

GLONASS

What GPS is to the US military, is GLONASS to the Russian military. The GLONASS space segment consists of nominally 24 satellites, organized in three orbital planes, with an inclination of 64.8° with the equator. Orbiting altitude is 19,130 km, with a period of revolution of 11 hours 16 min. GLONASS uses the PZ-90 as its reference system, and like GPS uses UTC as time reference, though with an offset for Russian daylight.

GLONASS radio signals are somewhat similar to that of GPS, but differ in the details. Satellites use different identifier schemes, and their navigation

message use other parameters. They also use different frequencies: GLONASS L1 is at approximately 1605 MHz (changes are underway), and L2 is at approximately 1248 MHz. Otherwise, the GLONASS system performance is rather comparable to that of GPS.

Galileo

In the 1990's, the European Union (EU) judged that it needed to have its own satellite-based positioning system, to become independent of the GPS monopoly and to support its own economic growth by providing services of high reliability under civilian control.

Galileo is the name of this EU system. The vision is that satellite-based positioning will become even bigger due to the emergence of mobile phones equipped with receivers, perhaps with some 400 million users by the year 2015. Development of the system has experienced substantial delays, and at the time of writing European ministers insist that Galileo should be up and running by the end of 2013. The completed system will have 27 satellites, with three in reserve, orbiting in one of three, equally spaced, circular orbits at an elevation of 23,222 km, inclined 56° with the equator. This higher inclination, when compared to that of GPS, has been chosen to provide better positioning coverage at high latitudes, such as northern Scandinavia where GPS performs rather poorly.

The Galileo Terrestrial Reference Frame (GTRF) will be a realization of the ITRS independently set up from that of GPS, so that one system can back-up for the other. Positional differences between the WGS84 and the GTRF will be at worst a few centimeters.

The Galileo System Time (GST) will closely follow International TAI Atomic Time (TAI) with a time offset of less than 50 nsec for 95 % of the time over any period of a year. Information on the actual offset between GST and TAI, and between GST and UTC (as used in GPS) will be broadcasted in the Galileo satellite signal.

Satellite-based augmentation systems

Satellite-based augmentation systems (SBAS) aim to improve accuracy and reliability of satellite-based positioning (see the section on network positioning, in support of safety-critical navigation applications such as

aircraft operations near airfields. The typical technique is to provide an extra, now geostationary, satellite that has a large service area like a continent, and which sends differential data about standard positioning satellites that are currently in view in its service area. If multiple ground reference stations are used, the quality of the differential data can be quite good and reliable. Signals typically use the frequency already in use by the positioning satellites, so that receivers can receive the differential code without problem.

Not all advantages of satellite augmentation will be useful for all receivers. For consumer market receivers, the biggest advantage, as compared to standard relative positioning, is that SBAS provides an ionospheric correction grid for its service area, from which a correction specific for the location of the receiver can be retrieved. This is not true in relative positioning, where the reference station determines the error it experiences, and simply broadcasts this information for nearby target receivers to use. With SBAS, the receiver obtains information that is best viewed as a geostatistical interpolation of errors from multiple reference stations. More advanced receivers will be able to deploy also other differential data such as corrections on satellite position and satellite clock drift.

Chapter 5

Data entry and preparation

Objectives:

1. To know the collection and use of data.
2. To prepare users of spatial data by drawing attention to issues concerning data accuracy and quality.
3. A range of procedures for data checking and clean-up.
4. Methods for interpolating point data.

Spatial data can be obtained from various sources. It can be collected from scratch, using direct spatial data acquisition techniques, or indirectly, by making use of existing spatial data collected by others. Under the first heading we could include field survey data and remotely sensed images. Under the second fall paper maps and existing digital data sets.

5.1 Spatial data input

5.1.1 Direct spatial data capture

One way to obtain spatial data is by *direct observation* of the relevant geographic phenomena. This can be done through ground-based field surveys, or by using remote sensors in satellites or airplanes.

Data which is captured directly from the environment is known as *primary data*.

With primary data the core concern in knowing its properties is to know the process by which it was captured, the parameters of any instruments used and the rigour with which quality requirements were observed.

Remotely sensed imagery is usually not fit for immediate use, as various

sources of error and distortion may have been present, and the imagery should first be freed from these. This is the domain of remote sensing.

An *image* refers to raw data produced by an electronic sensor, which are not pictorial, but arrays of digital numbers related to some property of an object or scene, such as the amount of reflected light.

For an image, no interpretation of reflectance values as thematic or geographic characteristics has taken place. When the reflectance values have been translated into some 'thematic' variable, we refer to it as a raster.

It is interesting to note that we refer to image *pixels* but to raster *cells*, although both are stored in a GIS in the same way.

In practice, it is not always feasible to obtain spatial data by direct spatial data capture. Factors of cost and available time may be a hindrance, or previous projects sometimes have acquired data that may fit the current project's purpose.

5.1.2 Indirect spatial data capture

In contrast to direct methods of data capture described above, spatial data can also be sourced *indirectly*. This includes data derived from existing paper maps.

Secondary data through scanning, data digitized from a satellite image, processed data purchased from data capture firms or international agencies, and so on. This type of data is known as *secondary data*:

Any data which is not captured directly from the environment is known as

Below we discuss key sources of secondary data and issues related to their use in analysis of which the user should be aware.

Digitizing

A traditional method data is through *digitizing* existing paper maps. This can be done using various techniques. Before adopting this approach, one must be aware that positional errors already in the paper map will further accumulate, and one must be willing to accept these errors.

There are two forms of digitizing: i) *on-tablet* and ii) *on-screen* manual digitizing. In on-tablet digitizing, the original map is fitted on a special surface (the tablet), while in on-screen digitizing, a scanned image of the map (or some other image) is shown on the computer screen.

The function of these points is to 'lock' a coordinate system onto the digitized data: the control points on the map have *known* coordinates, and by digitizing them we tell the system implicitly where all other digitized locations are. At least three control points are needed, but preferably more should be digitized to allow a check on the positional errors made.

These techniques are known as *semi-automatic* or *automatic* digitizing, depending on how much operator interaction is required. If vector data is to be distilled from this procedure, a process known as **vectorization** follows the scanning process. This procedure is less labour-intensive, but can only be applied

Scanning

An 'office' scanner illuminates a document and measures the intensity of the reflected light with a CCD array. The result is an image as a matrix of pixels, each of which holds an intensity value. Office scanners have a fixed maximum resolution, expressed as the highest number of pixels they can identify per inch;

Resolution the unit is dots-per-inch (dpi). For manual on-screen digitizing of a paper map, a resolution of 200–300 dpi is usually sufficient, depending on the thickness of the thinnest lines. For manual on-screen digitizing of aerial photographs, higher resolutions are recommended—typically, at least 800 dpi.

Semiautomatic digitizing requires a resolution that results in scanned lines of at least three pixels wide to enable the computer to trace the centre of the lines and thus avoid displacements. For paper maps, a resolution of 300–600 dpi is usually sufficient. Automatic or semi-automatic tracing from aerial photographs can only be done in a limited number of cases. Usually, the information from aerial photos is obtained through *visual interpretation*.

After scanning, the resulting image can be improved with various image processing techniques. It is important to understand that scanning does

not result in a structured data set of classified and coded objects. Additional work is required to recognize features and to associate categories and other thematic attributes with them.

Vectorization

The process of distilling points, lines and polygons from a scanned image is called *vectorization*. As scanned lines may be several pixels wide, they are often first thinned to retain only the centreline. The remaining centreline pixels are converted to series of (x, y) coordinate pairs, defining a polyline. Subsequently, OCR features are formed and attributes are attached to them. This process may be entirely automated or performed semi-automatically, with the assistance of an operator.

The phases of the vectorization process are illustrated in Figure 5.1.

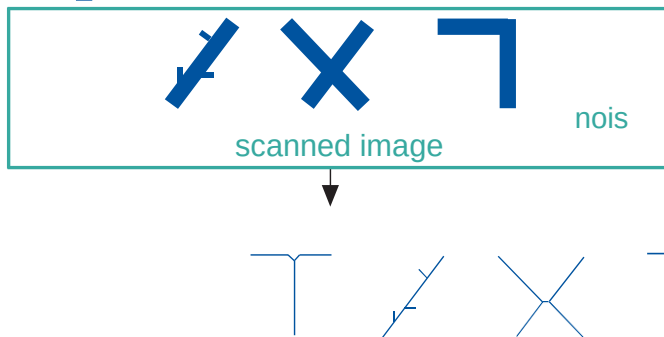


Fig. 5.1 phases of the vectorization process and the various sorts of small error caused by it. The post-processing phase makes the final repairs.

Selecting a digitizing technique

The choice of digitizing technique depends on the quality, complexity and contents of the input document. Complex images are better manually digitized; simple images are better automatically digitized. Images that are full of detail and symbols—like topographic maps and aerial photographs—are therefore better manually digitized.

In practice, the optimal choice may be a combination of methods. For example, contour line film separations can be automatically digitized and used to produce a DEM. Existing topographic maps must be digitized

manually, but new, geometrically corrected aerial photographs, with vector data from the topographic maps displayed directly over it, can be used for updating existing data files by means of manual on-screen digitizing.

5.1.2 Obtaining spatial data elsewhere

Over the past two decades, spatial data has been collected in digital form at increasing rate, stored in various databases by the individual producers for their own use and for commercial purposes. More and more of this data is being shared among GIS users. This is for several reasons. Some of this data is freely available, although other data is only available commercially, as is the case for most satellite imagery. High quality data remain both costly and time-consuming to collect and verify, as well as the fact that more and more GIS applications are looking at not just local, but national or even global processes. As we will see below, new technologies have played a key role in the increasing availability of geospatial data. As a result of this increasing availability, we have to be more careful that the data we have acquired is of sufficient *quality* to be used in analysis and decision making.

Clearinghouses and web portals

Spatial data can also be acquired from centralized repositories. More often those repositories are embedded in Spatial Data Infrastructures which make the data available through what is sometimes called a **spatial data clearinghouse**.

This is essentially a marketplace where data users can 'shop'. In Spatial Data Infrastructures will be no surprise that such markets for digital data have an entrance through the world wide web. The first entrance is typically formed by a web portal which categorizes all available data and provides a local search engine and links to data documentation (also called metadata). It often also points to data viewing and processing services. Standards-based geo-webservices have become the common technology behind such portal services.

Metadata

Metadata is defined as background information that describes all necessary information about the data itself. More generally, it is known as 'data about data'.

This includes:

- **Identification information:** Data source(s), time of acquisition, *etc.*
- **Data quality information:** Positional attribute and temporal accuracy, lineage, *etc.*
- **Entity and attribute information:** Related attributes, units of measure, *etc.*

In essence, metadata answer *who, what, when, where, why, and how* questions about all facets of the data made available. Maintaining metadata is an key part in maintaining data and information quality in GIS. This is because it can serve different purposes, from description of the data itself through to providing instructions for data handling. Depending on the type and amount of metadata provided, it could be used to determine the data sets that exist for a geographic location, evaluate whether a given data set meets a specified need, or to process and use a data set.

Data formats and standards

An important problem in any environment involved in digital data exchange is that of **data formats and data standards**. Different formats were implemented by different GIS vendors; different standards came about with different standardization committees. The phrase 'data standard' refers to an agreed upon ISO and OGC standards way of representing data in a system in terms of content, type and format. The good news about both formats and standards is that there are many to choose from; the bad news is that this can lead to a range of conversion problems. Several metadata standards for digital spatial data exist, including the International Organization for Standardization (ISO) and the Open Geospatial Consortium (OGC) standards.

5.2 Data quality

With the advent of satellite remote sensing, GPS and GIS technology, and the increasing availability of digital spatial data, resource managers and others who formerly relied on the surveying and mapping profession to supply high quality map products are now in a position to produce maps themselves. At the same time, GISs are being increasingly used for *decision support* applications, with Application requirements increasing reliance on secondary data sourced through data providers or via the internet, through geo-webservices. The implications of using low-quality data in important decisions are potentially severe. There is also a danger that uninformed GIS users introduce errors by incorrectly applying geometric and other transformations to the spatial data held in their database.

Discussion on positional, temporal and attribute accuracy, lineage, completeness, and logical consistency.

5.2.1 Accuracy and precision

So far we have used the terms error, accuracy and precision without appropriately defining them. Accuracy should not be confused with *precision*, which is a statement of the smallest unit of measurement to which data can be recorded. In conventional surveying and mapping practice, accuracy and precision are closely related. Instruments with an appropriate precision are employed, and surveying methods chosen, to meet specified accuracy tolerances.

In GIS, however, the numerical precision of computer processing and storage usually exceeds the accuracy of the data. This can give rise to so-called *spurious accuracy*, for example calculating area sizes to the nearest m² from coordinates obtained by digitizing a 1 : 50, 000 map.

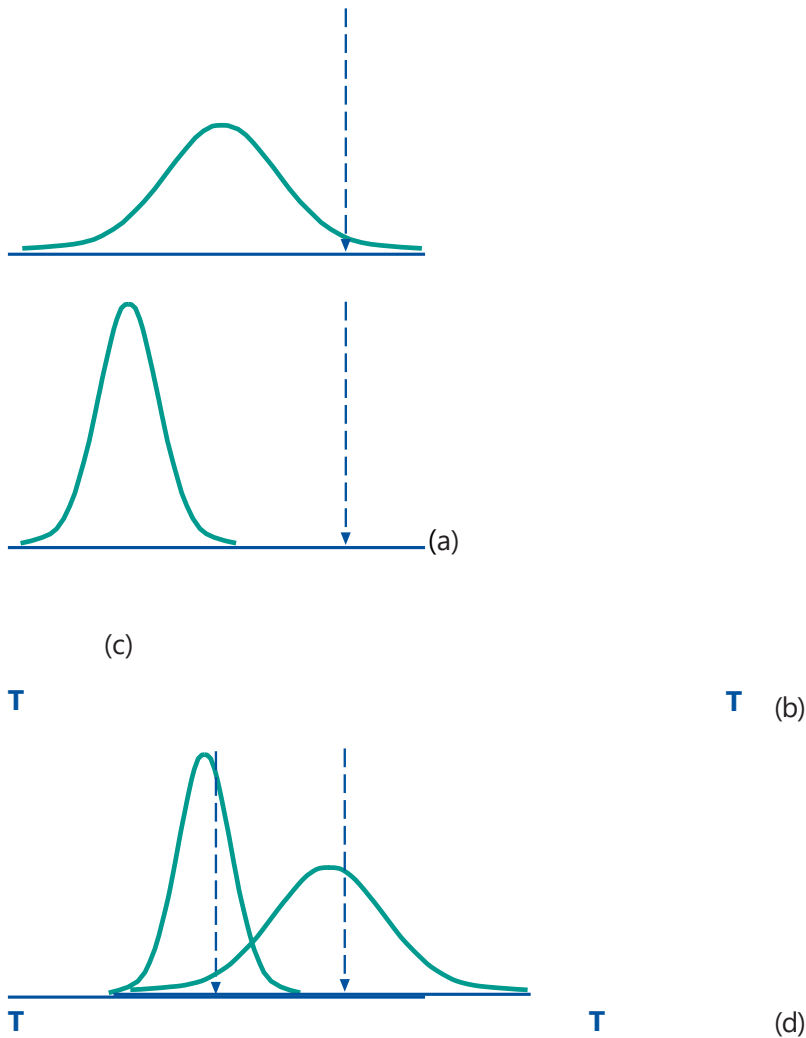


Figure 5.2: A measurement probability function and the underlying true value T:

(a) bad accuracy and precision, (b) bad accuracy/good precision,
(c) good accuracy/bad precision, and (d) good accuracy and precision.

5.2.2 Positional accuracy

The surveying and mapping profession has a long tradition of determining and minimizing errors. This applies particularly to land surveying and photogrammetry, both of which tend to regard positional and height errors as undesirable. Cartographers also strive to reduce geometric and

attribute errors in their products, and, in addition, define quality in specifically cartographic terms, for example quality of linework, layout, and clarity of text. It must be stressed that all measurements made with surveying and photogrammetric instruments are subject to error. These include:

- **Human errors** in measurement (e.g. reading errors) generally referred to as gross errors or *blunders*. These are usually large errors resulting from carelessness which could be avoided through careful observation, although it is never absolutely certain that all blunders have been avoided or eliminated.
- **Instrumental or systematic errors** (e.g. due to misadjustment of instruments). This leads to errors that vary systematically in sign and/or magnitude, but can go undetected by repeating the measurement with the same instrument. Systematic errors are particularly dangerous because they tend to accumulate.

So-called **random errors** caused by natural variations in the quantity being measured. These are effectively the errors that remain after blunders and systematic errors have been removed. They are usually small, and dealt with in least-squares adjustment.

Root mean square error

Location accuracy is normally measured as a *root mean square error (RMSE)*. The RMSE is similar to, but not to be confused with, the standard deviation of a statistical sample. The value of the RMSE is normally calculated from a set of check measurements (coordinate values from an independent source of higher accuracy for identical points). The differences at each point can be plotted as error vectors, as is done in Figure 5.3 for a single measurement. The error vector can be seen as having constituents in the x and y-directions, which can be recombined by vector addition to give the error vector representing its locational error.

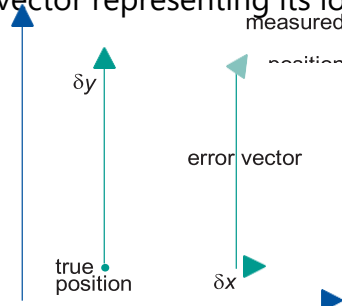


Figure 5.3: The positional error of a measurement can be expressed as a vector, which in turn can be viewed as the vector addition of its constituents in x and y -direction, respectively δx and δy

For each checkpoint, the error vector has components δx and δy . The observed errors should be checked for a *systematic* error component, which may indicate a (possibly repairable) lapse in the measurement method. Systematic error has occurred when $\sum \delta x \neq 0$ or $\sum \delta y \neq 0$.

The systematic error $\bar{\delta x}$ in x is then defined as the average deviation from the true value:

$$\bar{\delta x} = \frac{1}{n} \sum_{i=1}^n \delta x_i$$

Analogously to the calculation of the variance and standard deviation of a statistical sample, the root mean square errors m_x and m_y of a series of coordinate measurements are calculated as the square root of the average squared deviations:

$$m_x = \sqrt{\frac{1}{n} \sum_{i=1}^n \delta x_i^2} \text{ and } m_y = \sqrt{\frac{1}{n} \sum_{i=1}^n \delta y_i^2},$$

where δx^2 stands for $\delta x \cdot \delta x$. The total RMSE is obtained with the formula

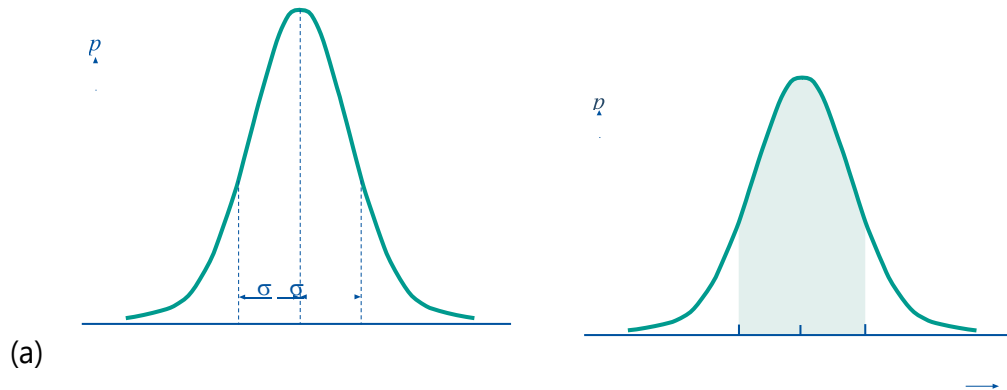
$$m_{\text{total}} = \sqrt{m_x^2 + m_y^2}$$

which, by the Pythagorean rule, is the length of the average (root squared) vector.

Accuracy tolerances

Many kinds of measurement can be naturally represented by a bell-shaped probability density function p , as depicted in Figure 5.4(a). This function is known as the **normal (or Gaussian) distribution** of a continuous, random variable, in the

figure indicated as Y . Its shape is determined by two parameters: μ , which is the mean expected value for Y , and σ which is the standard deviation of Y . A small σ leads to a more attenuated bell shape.



• **Figure 5.4:** (a) Probability density function p of a variable Y , with its mean μ and standard deviation σ

- (a) The probability that Y is in
(b) the range $[\mu - \sigma, \mu + \sigma]$

Any probability density function p has the characteristic that the area between its curve and the horizontal axis has size 1. Probabilities P can be inferred from p as the size of an area under p 's curve. Figure 5.4(b), for instance, depicts $P(\mu - \sigma \leq Y \leq \mu + \sigma)$, i.e. the probability that the value for Y is within distance σ from μ . In a normal distribution this specific probability for Y is always 0.6826.

The *RMSE* can be used to assess the probability that a particular set of measurements does not deviate too much from, i.e. is within a certain range of, the 'true' value. In the case of coordinates, the probability density function often is considered to be that of a two-dimensional normally distributed variable (see Figure 5.5). The three standard probability values associated with this distribution are:

0.50 for a circle with a radius of $1.1774 m_x$ around the mean (known as the **circular error probable, CEP**);

0.6321 for a circle with a radius of $1.412 m_x$ around the mean (known as the **root mean square error, RMSE**);

0.90 for a circle with a radius of $2.146 m_x$ around the mean (known as the **circular map accuracy standard, CMAS**).

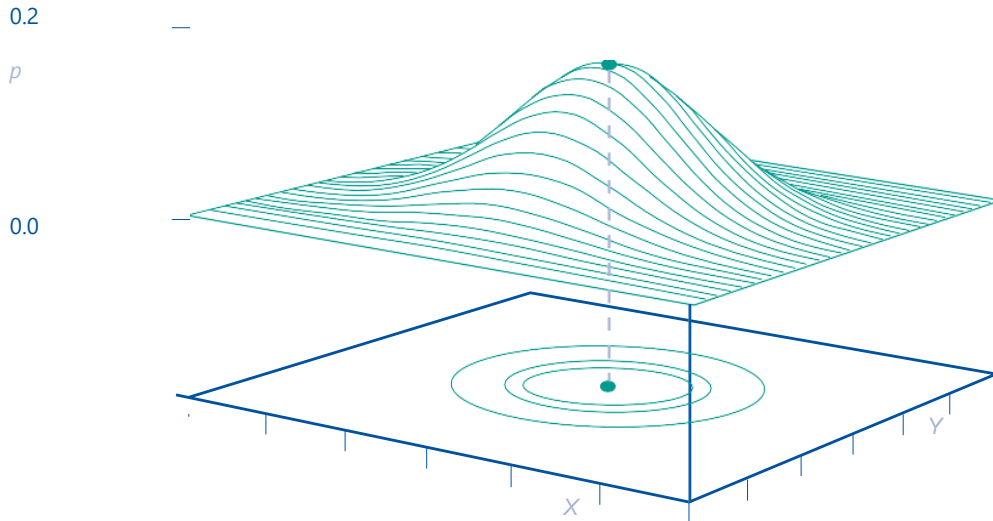


Figure 5.5: Probability density p of a normally distributed, two-dimensional variable (X, Y) (also known as a normal, bivariate distribution).

In the ground plane, from inside out, are indicated the circles respectively associated with CEP, RMSE and CMAS.

The RMSE provides an estimate of the spread of a series of measurements around their (assumed) 'true' values. It is therefore commonly used to assess the quality of transformations such as the absolute orientation of photogrammetric models or the spatial referencing of satellite imagery. The RMSE also forms the basis of various statements for reporting and verifying compliance with defined map accuracy *tolerances*. An example is the American National Map Accuracy Standard, which states that:

"No more than 10% of well-defined points on maps of 1:20,000 scale or greater may be in error by more than 1/30 inch."

Normally, compliance to this tolerance is based on at least 20 well-defined checkpoints.

The epsilon band

As a line is composed of an infinite number of points, confidence limits can be described by a so-called epsilon (ϵ) or Perkal band at a fixed distance on either side of the line (Figure 5.6). The width of the band is based on an estimate of the probable location error of the line, for example to reflect the accuracy of manual digitizing. The epsilon band may be used as a simple means for assessing the likelihood that a point receives the correct attribute value (Figure 5.7).

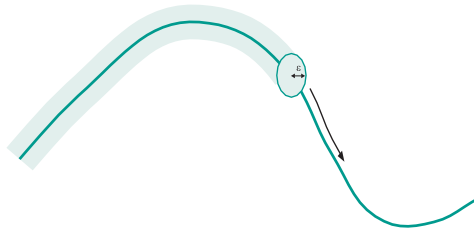


Figure 5.6: The ϵ or Perkal band is formed by rolling an imaginary circle of a given radius along a line

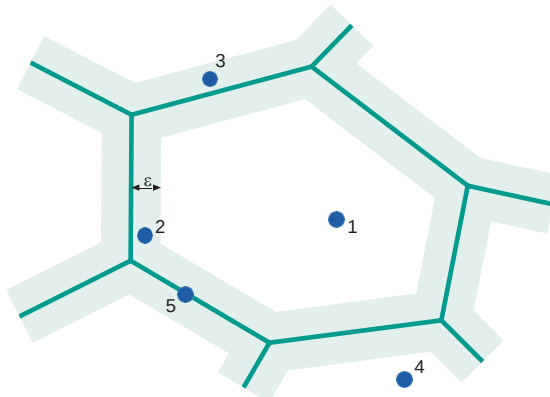


Figure 5.7: The ϵ -band may be used to assess the likelihood that a point falls within a particular polygon. Source: [43]. Point 3 is less likely part of the middle polygon than point

Describing natural uncertainty in spatial data

There are many situations, particularly in surveys of natural resources, where, according to Burrough, “practical scientists, faced with the problem of dividing up undividable complex continua have often imposed their own crisp structures on the raw data”. In practice, the results of classification are normally combined with other categorical layers and continuous field data to identify, for example, areas suitable for a particular land use. In a GIS, this is normally achieved by overlaying the appropriate layers using logical operators.

Particularly in natural resource maps, the boundaries between units may not actually exist as lines but only as transition zones, across which one area continuously merges into another. In these circumstances, rigid measures of positional accuracy, such as RMSE (Figure 5.3), may be virtually insignificant in comparison to the uncertainty inherent in vegetation and soil boundaries, for example.

In conventional applications of the error matrix to assess the quality of nominal (categorical) data such as land use, individual samples can be considered in terms of Boolean set theory. The Boolean *membership function* is binary, i.e. an element is either member of the set (membership is true) or it is not member of the set (membership is false). Such a membership notion is well-suited to the description of spatial features such as land parcels where no ambiguity is involved and an individual ground truth sample can be judged to be either correct or incorrect.

In GIS, fuzzy set theory appears to have two particular benefits:

1. The ability to handle logical modelling (map overlay) operations on inexact data, and
2. The possibility of using a variety of natural language expressions to qualify uncertainty.

5.2.3 Attribute accuracy



We can identify two types of attribute accuracies. These relate to the type of data we are dealing with:

For *nominal or categorical* data, the accuracy of labeling (for example the type of land cover, road surface, etc).

For *numerical* data, numerical accuracy (such as the concentration of pollutants in the soil, height of trees in forests, etc).

It follows that depending on the data type, assessment of attribute accuracy may range from a simple check on the labelling of features—for example, is a road classified as a metalled road actually surfaced or not?—to complex statistical procedures for assessing the accuracy of numerical data, such as the percentage of pollutants present in the soil.

When spatial data are collected in the field, it is relatively easy to check on the appropriate feature labels. In the case of remotely sensed data, however, considerable effort may be required to assess the accuracy of the classification procedures. This is usually done by means of checks at a number of sample points.

Classified image	Reference data			total
	Forest	Agriculture	Urban	
Forest	62	5	0	67
Agriculture	2	18	0	20
Urban	0	1	12	13
total	64	24	12	100

Table 5.1: Example of a simple error matrix for assessing map attribute accuracy.

The overall accuracy is $(62 + 18 + 12) / 100 = 92\%$

5.2.4 Temporal accuracy

As noted, the amount of spatial data sets and archived remotely sensed data has increased enormously over the last decade. These data can provide useful temporal information such as changes in land ownership and the monitoring of environmental processes such as deforestation. Analogous to its positional and attribute components, the quality of spatial data may also be assessed in terms of its *temporal accuracy*. For a static feature this refers to the difference in the values of its coordinates at two different times.

This includes not only the accuracy and precision of time measurements (for example, the date of a survey), but also the temporal consistency of different data sets. Because the positional and attribute components of spatial data may change together or independently, it is also necessary to consider their temporal validity. For example, the boundaries of a land parcel may remain fixed over a period of many years whereas the ownership attribute may change more frequently.

5.2.5 Lineage

Lineage describes the history of a data set. In the case of published maps, some lineage information may be provided as part of the metadata, in the form of a note on the data sources and procedures used in the compilation of the data. Examples include the date and scale of aerial photography, and the date of field verification. Especially for digital data sets, however, lineage may be defined more formally as:

"that part of the data quality statement that contains information that describes the source of observations or materials, data acquisition and compilation methods, conversions, transformations, analyses and derivations that the data has been subjected to, and the assumptions and criteria applied at any stage of its life." [14]

All of these aspects affect other aspects of quality, such as positional accuracy. Clearly, if no lineage information is available, it is not possible to adequately evaluate the quality of a data set in terms of 'fitness for use'.

5.2.6 Completeness

Completeness refers to whether there are data lacking in the database compared to what exists in the real world. Essentially, it is important to be able to assess what does and what does not belong to a *complete* dataset as intended by its producer.

It might be incomplete (i.e. it is 'missing' features which exist in the real world), or overcomplete (i.e. it contains 'extra' features which do not belong within the scope of the data set as it is defined).

5.2.7 Logical consistency

For any particular application, (predefined) logical rules concern:

The *compatibility* of data with other data in a data set (e.g. in terms of data format),

- The absence of any ***contradictions*** within a data set,
- The ***topological consistency*** of the data set, and

The allowed attribute *value ranges*, as well as combinations of attributes. For example, attribute values for population, area, and population density must agree for all entities in the database.

The absence of any inconsistencies does not necessarily imply that the data are accurate.

5.3 Data preparation

Spatial data preparation aims to make the acquired spatial data fit for use. Images may require enhancements and corrections of the classification scheme of the data. Vector data also may require editing, such as the trimming of overshoots of lines at intersections, deleting duplicate lines, closing gaps in lines, and generating polygons. Data may require conversion to either vector format or raster format to match other data sets which will be used in the analysis. Additionally, the data preparation process includes associating attribute data with the spatial features through either manual input or reading digital attribute files into the GIS/DBMS.

Intended use

The intended use of the acquired spatial data may require only a subset of the original data set, as only some of the features are relevant for subsequent analysis or subsequent map production. In these cases, data and/or cartographic generalization can be performed on the original data set.

5.3.1 Data checks and repairs

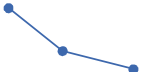
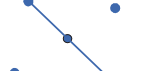
Acquired data sets must be checked for quality in terms of the accuracy, consistency and completeness parameters discussed above. Often, errors can be identified automatically, after which manual editing methods can be applied to correct the errors. Alternatively, some software may identify and automatically correct certain types of errors. Below, we focus on the geometric, topological, and attribute components of spatial data.

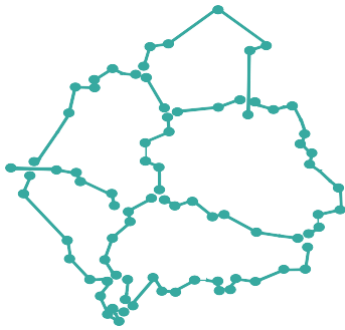
‘Clean-up’ operations are often performed in a standard sequence. For example, crossing lines are split before dangling lines are erased, and nodes are created at intersections before polygons are generated. These are illustrated in Table 5.2.

With polygon data, one usually starts with many polylines, in an unwieldy format known as spaghetti data, that are combined in the first step (from Figure 5.9(a) to (b)). This results in fewer polylines with more internal vertices. Then, polygons can be identified (c). Sometimes, polylines that should connect to form closed boundaries do not, and therefore must be connected (either manually or automatically); this step is not indicated in the figure. In

a final step, the elementary topology of the polygons can be derived (d).

Table 5.2: Clean-up operations for vector data

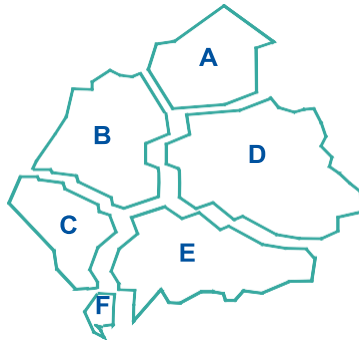
Before cleanup	After cleanup	Description	Before cleanup	After cleanup	Description
		Erase duplicates or sliver lines			Extend undershoots
		Erase short objects			Snap clustered nodes
		Break crossing objects			Erase dangling objects or overshoots
		Dissolve polygons			Dissolve nodes into vertices



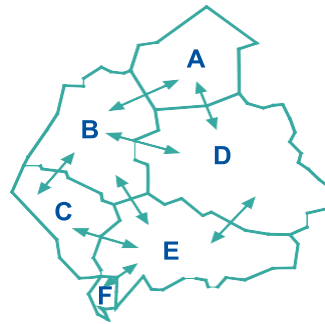
(a) Spaghetti data



(b) Spaghetti data (cleaned)



(c) Polygons



(d) Topology

Figure 5.9: Successive clean-up operations for vector data, turning spaghetti data into topological structure.

Associating attributes

Attributes may be automatically associated with features that have unique identifiers. We have already discussed these techniques in Section 3.5. In the case of vector data, attributes are assigned directly to the features, while in a raster the attributes are assigned to all cells that represent a feature.

Rasterization or vectorization

Vectorization produces a vector data set from a raster. We have looked at this in some sense already: namely in the production of a vector set from a scanned image. Another form of vectorization takes place when we want to identify features or patterns in remotely sensed imagery.

If much or all of the subsequent spatial data analysis is to be carried out on raster data, one may want to convert vector data sets to raster data. This process is known as ***rasterization***.

It involves assigning point, line and polygon attribute values to raster cells that overlap with the respective point, line or polygon. To avoid information loss, the raster resolution should be carefully chosen on the basis of the geometric resolution. A cell size which is too large may result in cells that cover parts of multiple vector features, and then ambiguity arises as to what value to assign to the cell. If, on the other hand, the cell size is too small, the file size of the raster may increase significantly.

Rasterization itself could be seen as a 'backwards step': firstly, raster boundaries are only an *approximation* of the objects' original boundary. Secondly, the original 'objects' can no longer be treated as such, as they have lost their topological properties. Often the reason for rasterisation is because it facilitates easier combination with other data sources also in raster formats, and/or because there are several analytical techniques which are easier to perform upon raster data and generating raster data from them when needed. Obviously, the issue of performance trade-off must be looked into.

Topology generation

We have already discussed derivation of topology from vectorized data sources. However, more topological relations may sometimes be needed, for instance in networks, e.g. the questions of line connectivity, flow direction, and which lines have overland underpasses. For polygons, questions that may arise involve What kind of topology is polygon incursion.

5.2.2 Combining data from multiple sources

A GIS project usually involves multiple data sets, so the next step addresses the issue of how these multiple sets relate to each other. There are four fundamental cases to be considered in the combination of data from different sources:

1. They may be about the same area, but differ in *accuracy*,
2. They may be about the same area, but differ in choice of *representation*,
3. They may be about adjacent areas, and have to be *merged* into a single data set.
4. They may be about the same or adjacent areas, but referenced in different *coordinate systems*.

We look at these situations below. They are best understood with an example.

Other data preparation functions

A range of other data preparation functions exist that support conversion or adjustment of the acquired data to format requirements that have been defined for data storage purposes. These include:

Format transformation functions. These convert between data formats of different systems or representations, e.g. reading a DXF file into a GIS. Although we will not focus on the technicalities here, the user should be warned that *conversions from one format to another may cause problems*. The reason is that not all formats can capture the same information, and therefore conversions often mean loss of information. If one obtains a spatial data set in format *F*, but needs it in format *G* (for instance because the locally preferred GIS package requires it), then usually a conversion function can be found, often within the same GIS software package. The key to successful conversion is to also find an inverse conversion, back from *G* to *F*, and to ascertain whether the double conversion back to *F* results in the same data set as the original. If this is the case, both conversions are not causing information loss, and can safely be applied.

Graphic element editing. Manual editing of digitized features so as to correct errors, and to prepare a clean data set for topology building.

Coordinate thinning. A process that is often applied to remove redundant or excess vertices from line representations, as obtained from digitizing.

5.4 Point data transformation

We may want to transform our points into other representations in order to facilitate interpretation and/or integration with other data. Examples include defining homogeneous areas (polygons) from our point data, or deriving contour lines. This is generally referred to as *interpolation*, i.e. the calculation of a

Interpolation value from 'surrounding' observations. The principle of spatial autocorrelation plays a central part in the process of interpolation

In order to predict the value of a point for a given (x, y) location, we could simply find the 'nearest' known value to the point, and assign that value. This is the simplest form of interpolation, known as *nearest-neighbour* interpolation. We might instead choose to use the distance that points are away from (x, y) to weight their importance in our calculation.

Video 2 - GIS Interpolation Type Comparison

Source:<https://www.youtube.com/watch?v=to6Eufi58hM>

In some instances we may be dealing with a data type that limits the type of interpolation we can do (refer to page 75 for a brief background). A fundamental issue in this respect is what kind of phenomena we are considering: is it a **discrete field**—such as geological units, for instance—in which the values are of a qualitative nature and the data is categorical, or is it a **continuous field**—like elevation, temperature, or salinity—in which the values are of a quantitative nature, and represented as continuous measurements? This distinction matters because we are limited to nearest-neighbour interpolation for discrete data.

A simple example is given in Figure 5.13. Our field survey has taken only two measurements, one at P and one at Q. The values obtained in these two locations are represented by a dark and light green tint, respectively. If we are dealing with qualitative data, and we have no further knowledge, the only assumption we can make for other locations is that those nearer to P probably have P's value, whereas those nearer to Q have Q's value. This is illustrated in part (a).

If, on the contrary, our field is quantitative, we can let the values of P and Q both contribute to values for other locations. This is done in part (b) of the figure. To what extent the measurements contribute is determined by the interpolation function. In the figure, the contribution is expressed in terms of the ratio of distances to P and Q. We will see in the sequel that the choice of interpolation function is a crucial factor in any method of point data transformation.

How we represent a field constructed from point measurements in the GIS also depends on the above distinction. A discrete field can either be represented as a classified raster or as a polygon data layer, in which each polygon has been assigned a (constant) field value. A continuous field can be represented as an unclassified raster, as an isoline (thus, vector) data layer, or perhaps as a TIN. Some GIS software only provide the option of generating raster output, requiring an intermediate step of raster to vector conversion. The choice of representation depends on what will be done with the data in the analysis phase.



Figure 5.13: A geographic field representation obtained from two point measurements: (a) for qualitative (categorical), and (b) for quantitative (continuous) point measurements. The value measured at *P* is represented as dark green at *Q* as light green.

5.4.1 Interpolating discrete data

If we are dealing with discrete (nominal, categorical or ordinal) data, we are effectively restricted to using nearest-neighbour interpolation. This is the situation shown in Figure 5.13(a), though usually we would have many more points.

In a nearest-neighbour interpolation, each location is assigned the value of the closest measured point. Effectively, this technique will construct 'zones' around the points of measurement, with each point belonging to a zone assigned the same value. Effectively, this represents an assignment of an existing value (or category) to a location.

If the desired output was a polygon layer, we could construct *Thiessen polygons* around the points of measurement. The boundaries of such polygons, by definition, are the locations for which more than one point of measurement is the closest point. An illustration is provided in Figure 5.14.

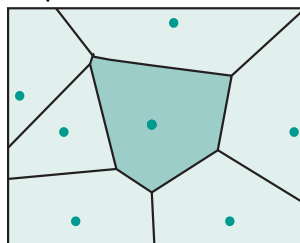


Figure 5.14: Generation of Thiessen polygons for qualitative point measurements. The measured points are indicated in dark green; the darker area indicates all locations assigned with the measurement value of the central point

5.4.2 Interpolating continuous data

Interpolation of values from continuous measurements is significantly more complex. This is the situation of Figure 5.13(b), but again, usually with many more point measurements.

Since the data are continuous, we can make use of measured values for interpolation. There are many continuous geographic fields—elevation, temperature and ground water salinity are just a few examples. Commonly, continuous fields are represented as rasters, and we will almost by default assume that they are. Alternatives exist though, as we have seen in discussions in Chapter 2. The main alternative for continuous field representation is a polyline vector layer, in which the lines are isolines. We will also address these issues of representation below.

The aim is to use measurements to obtain a representation of the entire field using point samples. In this section we outline four techniques to do so:

1. Trend surface fitting using *regression*,
2. Triangulation,
3. Spatial moving averages using *inverse distance weighting*,
4. Kriging.

1. Trend surface fitting

In trend surface fitting, the assumption is that the entire study area can be represented by a formula $f(x, y)$ that for a given location with coordinates (x, y) will give us the approximated value of the field in that location.

The key objective in trend surface fitting is to derive a formula that best describes the field. Various classes of formulæ exist, with the simplest being the one that describes a flat, but tilted plane:

$$f(x, y) = c_1 \cdot x + c_2 \cdot y + c_3.$$

If we believe—and this judgement must be based on domain expertise—that the field under consideration can be best approximated by a tilted plane, then the problem of finding the best plane is the problem of determining best values for the coefficients c_1 , c_2 and c_3 . This is where the point measurements earlier obtained become important. Statistical techniques known as *regression techniques*.

Regression can be used to determine values for these coefficients c_i that best fit with the measurements. A plane will be fitted through the measurements that makes the smallest overall error with respect to the original measurements.

In Figure 5.15, we have used the same set of point measurements, with four different approximation functions. Part (a) has been determined under the assumption that the field can be approximated by a tilted plane, in this case with a downward slope to the southeast. The values found by regression techniques were: $c_1 = -1.83934$, $c_2 = 1.61645$ and $c_3 = 70.8782$, giving us:

$$f(x, y) = -1.83934 \cdot x + 1.61645 \cdot y + 70.8782.$$

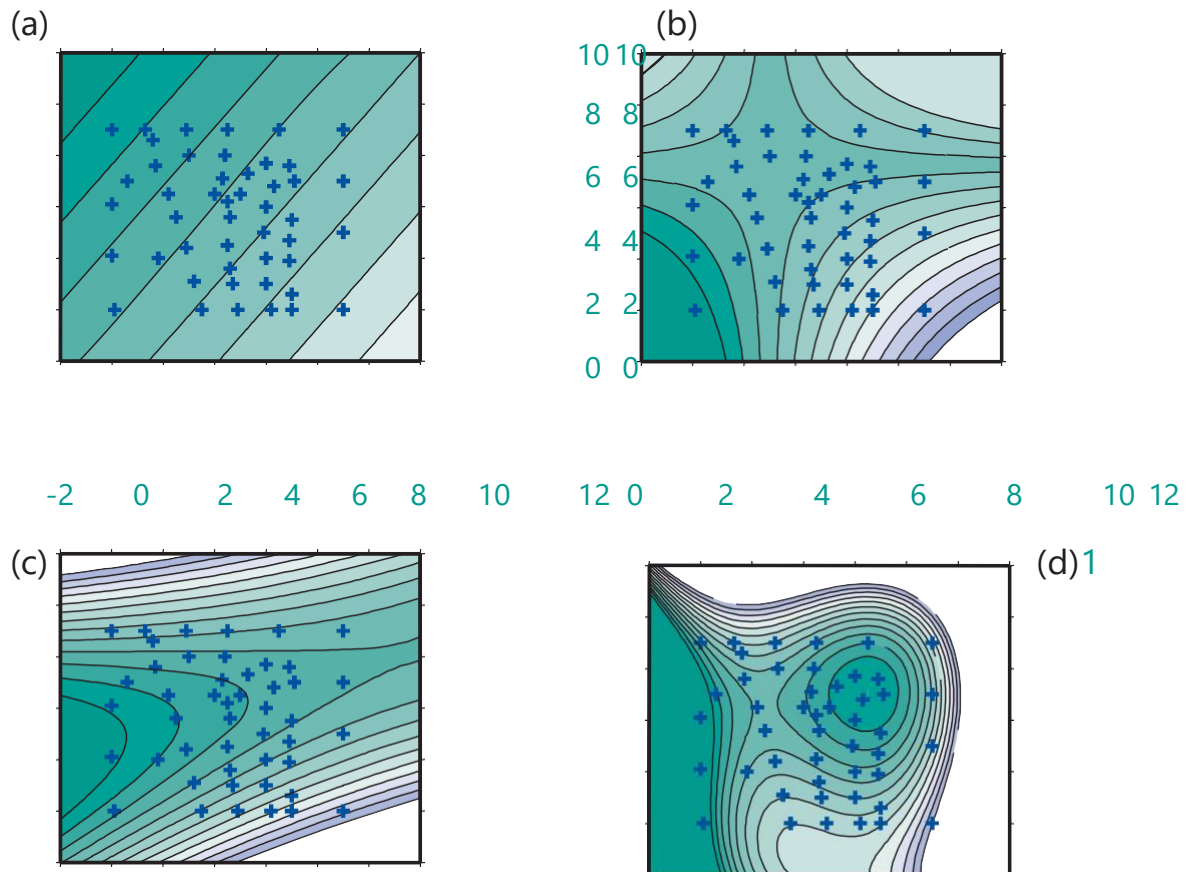


Figure 5.15: Various Clearly, not all fields are representable as simple, tilted planes. Sometimes, the theory of the application domain will dictate that the best approximation of the field is a more complicated, higher-order polynomial function. Three such functions were the basis for the fields illustrated in Figure 5.15(b)–(d).

The simplest extension from a tilted plane, that of *bilinear saddle*, expresses some dependency between the x and y dimensions:

$$f(x, y) = c_1 \cdot x + c_2 \cdot y + c_3 \cdot xy + c_4.$$

This is illustrated in part (b). A further step up the ladder of complexity is to consider *quadratic surfaces*, described by:

$$f(x, y) = c_1 \cdot x^2 + c_2 \cdot x + c_3 \cdot y^2 + c_4 \cdot y + c_5 \cdot xy + c_6.$$

The objective is to find six values for our coefficients that best match with the measurements. A bilinear saddle and a quadratic surface have been fitted through our measurements in Figure 5.15(b) and (c), respectively.

Part (d) of the figure illustrates the most complex formula of the surfaces in Figure 5.15, the *cubic surface*. It is characterized by the following formula:

$$f(x, y) = c_1 \cdot x^3 + c_2 \cdot x^2 + c_3 \cdot x + c_4 \cdot y^3 + c_5 \cdot y^2 + c_6 \cdot y + c_7 \cdot x^2 y + c_8 \cdot xy^2 + c_9 \cdot xy + c_{10}$$

The regression techniques applied for Figure 5.15 determined the following values for the coefficients c_i :

Fig 5.15	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10
(a)	-1.83934	1.61645	70.8782							
(b)	-5.61587	-2.95355	0.993638	89.0418						
(c)	0.000921084	-5.02674	-1.34779	7.23557	0.813177	76.9177				
(d)	-0.473086	6.88096	31.5966	-0.233619	1.48351	-2.52571	-0.115743	-0.052568	2.16927	96.8207

Trend surface fitting is a useful technique of continuous field approximation, though determining the 'best fit' values for the coefficients c_i is a time-consuming operation, especially with many point measurements. Once these best values have been determined, we know the formula, making it possible to compute an approximated value for any location in the study area.

It is possible to use trend surfaces for both *global* and *local* trends. Global trend surface fitting is based on the assumption that the entire study area can be approximated by the same mathematical surface. However in many cases, the

Local trend surface fitting is not a popular technique in practical applications, because they are relatively difficult to implement, and other techniques such as moving windows are better for the representation and identification of local trends.

If we know the polynomial, it is relatively simple to generate a raster layer, given an appropriate cell resolution and an approximation function for the cell's value.

In some cases it is more accurate to assign the average of the computed values for all of the cell's corner points. In order to generate a vector layer representing this

Generating trend surfaces data, *isolines* can be derived, for a given set of intervals. The specific techniques of generating isolines are not discussed here, however, triangulation techniques discussed below can play a role.

2. Triangulation

Another way of interpolating point measurements is by triangulation. Essentially, this technique constructs a triangulation of the study area from the known measurement points. Preferably, the triangulation should be a *Delaunay triangulation*.³ After having obtained it, we may define for which values of the field we want to construct isolines. For instance, for elevation, we might want to have the 100 m-isoline, the 200 m-isoline, and so on. For each edge of a triangle, a geometric computation can be performed that indicates which isolines intersect it, and at what positions they do so. A list of computed locations, all at the same field value, is used by the GIS to construct the isoline. This is illustrated in Figure 5.16.

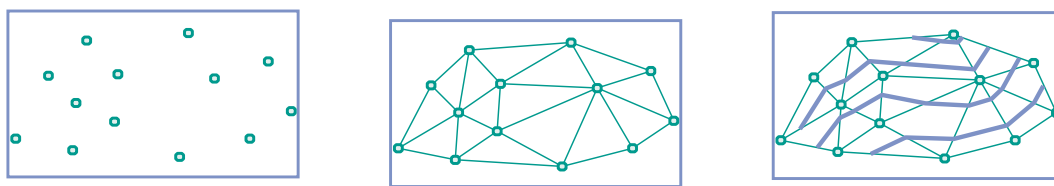
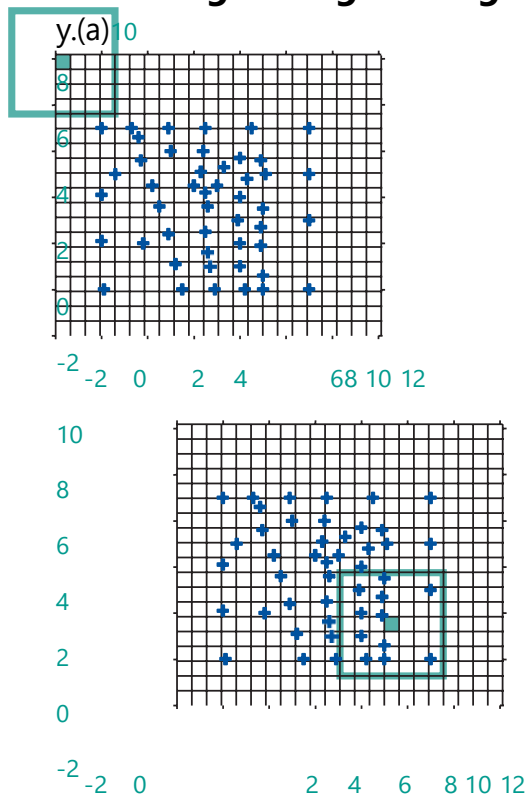


Figure 5.16: Triangulation as a means of interpolation. (a) known point measurements; (b) constructed triangulation on known points; (c) isolines constructed from the triangulation.

3. Moving averages using inverse distance weighting (IDW)



Moving window averaging attempts to directly derive a raster dataset from a set of sample points. This is why it is sometimes also called 'gridding'. The principle behind this technique is illustrated in Figure 5.17. The cell values for the output raster are computed one by one. To achieve this, a 'window' (also known as a kernel) is defined, and initially placed over the top left raster cell. Measurement Moving window averaging points falling inside the window contribute to the averaging computation, those outside the window do not. This is why moving window averaging is said to be a *local* interpolation method. After the cell value is computed and assigned to the cell, the window is moved one cell to the right, and the computations are performed for that cell. Successively, all cells of the raster are visited in this way as an example of moving window averaging. In blue, the measurement points. A virtual window is moved over the raster cells one by one, and some averaging function computes a field value for the cell, using measurements within the window

In part (b) of the figure, the 295th cell value out of the 418 in total, is being computed. This computation is based on eleven measurements, while that of the first cell had no measurements available. Where this is the case, the cell should be assigned a value that signals this 'non-availability of measurements'.

Suppose there are n measurements selected in a window, and that a measurement is denoted as m_i . The simplest averaging function will compute the arithmetic mean, treating all measurements equally:

$$\frac{1}{n} \sum_{i=1}^n m_i$$

The principle of spatial autocorrelation suggests that measurements closer to the cell centre should have greater influence on the predicted value than those further away. In order to account for this, a distance factor can be brought into the averaging function. Functions that do this are called *inverse distance weighting functions* (IDW). This is one of the most commonly used functions in interpolating spatial data.

Let us assume that the distance from measurement point i to the cell centre is denoted by d_i . Commonly, the weight factor applied in inverse distance weighting is the distance squared, but in the general case the formula is:

$$\frac{\sum_{i=1}^n m_i}{\sum_{i=1}^n \frac{1}{d_i^p}}$$

Moving window averaging has many parameters. As experimentation with any GIS package will demonstrate, picking the right parameter settings may make quite a difference for the resulting raster. We discuss some key parameters below. **Raster resolution:** Too large a cell

size will smooth the function too much, removing local variations; too small a cell size will result in large clusters of equally valued cells, with little added value.

Shape/size of window: Most procedures use square windows, but rectangular, circular or elliptical windows are also possible. These can be useful in cases where the measurement points are distributed regularly at fixed distance over the study area, and the window shape must be chosen to ensure that each raster cell will have its window include the same number of measurement points. The size of the window is another important matter. Small windows tend to exaggerate local extreme values, while large windows have a smoothing effect on the predicted field values.

Selection criteria: Not necessarily all measurements within the window need to be used in averaging. We may choose to select use at most five, (nearest) measurements, or we may choose to only generate a field value if more than three measurements are in the window.⁴

Averaging function: A final choice is which function is applied to the selected measurements within the window. It is possible to use different distance-weighting functions, each of which will influence the calculation of the resulting value.

4. Kriging

Kriging was originally developed by mining geologists attempting to derive accurate estimates of mineral deposits in a given area from limited sample measurements. It is an advanced interpolation technique belonging to the field of *geostatistics*, which can deliver good results if applied properly and with enough sample points. Kriging is usually used when the variation of an attribute and/or the density of sample points is such that simple methods of interpolation may give unreliable predictions.

Kriging is based on the notion that the spatial change of a variable can be described as a function of the distance between points. It is similar to IDW interpolation, in that it the surrounding values are weighted to derive a value for an unmeasured location. However, the kriging method also looks at the overall spatial arrangement of the measured points and the spatial correlation between their values, to derive values for an unmeasured location.

The first step in the kriging procedure is to compare successive pairs of point measurements to generate a *semi-variogram*. In the second step, the semi-vario-gram is used to calculate the weights used in interpolation.

Although kriging is *Semi-variogram* a powerful technique, it should not be applied without a good understanding of geostatistics, including the principle of spatial autocorrelation. For more detail on the various kriging methods, readers are referred to [11].

Graded Questions

Chapter 04

Spatial referencing and positioning

1. Explain the spatial referencing surfaces for mapping.
2. Write short note on Geoid and Ellipsoid.
3. Explain local horizontal and vertical datum.
4. Describe Triangulation network with the help of diagram.
5. Explain the 2D and 3D Geographic coordinate system.
6. What is Map projection? Explain the types of map projection with the help of diagrams in brief.
7. Explain the coordinate transformation with the help of 2D polar to 2D Cartesian.
8. Write short note on datum transformation.
9. What is map projection?
10. Describe the classification of map projection.
11. What is Satellite based positioning? Explain.
12. What is absolute positioning?
13. What are the errors in absolute positioning?
14. Write short note on
15. Errors related in space segment
16. Errors in related to medium
17. Errors related to the receiver's environment
18. Errors related to relative geometry of satellites and receiver
19. Explain positioning technology and describe GPS, GLONASS and Galileo in brief.
20. Assume that the semi-major axis a of an ellipsoid is 6378137 m and the flattening f is 1:298.257. Using these facts determine the semi-minor axis b (make use of the given equations).
21. You are required to match GPS data with some map data. The GPS data and the map layer are based on different horizontal datums. Which steps should you take to make the GPS data spatially compatible with the map data?

Chapter 05

Data entry and preparation

1. Rasterization of vector data is sometimes required in data preparation. What reasons may exist for this? If it is needed, the raster resolution must be carefully selected. Argue why.
2. Write a short note on direct and indirect spatial data capture.
3. Explain the following terms i) digitizing ii) scanning iii) vectorization
4. What is digitizing? What are the types of digitizing? Explain.
5. Explain vectorization in brief.
6. What is scanning?
7. What is metadata? Explain dataformats and standards.
8. Write short notes on following terms in concerns with data quality.
 - Accuracy and Precision
 - Positional accuracy
 - Attribute accuracy.
 - Temporal accuracy
 - Lineage
 - Logical consistency
9. Explain Root Mean Square Error (RMSE).
10. Describe natural uncertainty in spatial data.
11. How do we data checks and repairing in data preparation process?
12. Write short note on
13. Explain point data transformation.
14. Explain interpolating discrete data.
15. Explain interpolating continuous data.
16. Write short note on Kriging.
17. Write short note on triangulation.
18. Write short note on Trend surface fitting.

MULTIPLE CHOICE QUESTIONS FOR QUIZ

1. What is the NAVSTAR GPS?
 - a. Global Positioning System**
 - b. Geographic Projection System
 - c. General Policy System
 - d. Geographic Portion Service

2. Which Time system is used for satellite positioning
 - a. Atomic Clock
 - b. UTI
 - c. UTC (Universal Time Coordinate)**
 - d. UTM

3. Errors related to the space segment is due to _____
 - a. Incorrect clock reading and orbit positioning**
 - b. Ionosphere
 - c. Incorrect medium
 - d. Incorrect signal receivers

4. GPS and GLONASS and Galileo are the _____
 - a. Environmental Technology
 - b. Positional Technology**
 - c. Fundamental Technology
 - d. Advanced Technology

5. Galileo is the name of the _____
 - a. Environmental unit
 - b. European Union (EU)**
 - c. Geographic Technology
 - d. Advanced Technology

6. Data which is captured directly from the environment is known as _____

a. Secondary data

b. Primary data

c. Historical data

d. Important data

7. _____ digitizing type in which the original map is fitted on a special surface

a. On tablet

b. On screen

c. On scanner

d. On page

8. The process of distilling points, lines and polygons from a scanned image is called _____

a. Rasterization

b. Vectorization

c. Transformation

d. Conversion

9. As a line is composed of an infinite number of points, confidence limits can be described by a so-called _____ at a fixed distance on either side of the line.

a. epsilon (") or Perkal band

b. gipsililon

c. point_line

d. spatial-on

10. _____ describes the history of a data set.

a. Linage

b. signage

c. oldage

d. Historical data

e. Important data

11. _____ involves checking for errors, inconsistencies, and simplification and merging existing spatial data sets
- a. Vectorization and Rasterization
 - b. Data transformation
 - c. Data cleaning and preparation**
 - d. Data Conversion
12. Which one is not the part of interpolating continuous data.
- a. Attribute accuracy**
 - b. Kriging
 - c. Triangulation
 - d. Trend surface fitting
13. _____ is based on the notion that the spatial change of a variable can be described as a function of the distance between points
- a. Attribute accuracy
 - b. Kriging**
 - c. Triangulation
 - d. Trend surface fitting
14. _____ converts vector data sets to raster data.
- a. Rasterization**
 - b. Vectorization
 - c. Transformation
 - d. Conversion
15. Data which is captured indirectly from the environment is known as _____
- a. Secondary data**
 - b. Primary data
 - c. Historical data
 - d. Important data